

A PRACTICAL SYSTEMS GUIDE

Building a Coding Agent Harness

From Model Calls to a Safe, Observable, Evaluable Runtime

A practical systems guide built with Rust and proven through Quecto

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Preface

You will not learn much by reading this book about models. You will learn a great deal by reading it about harnesses.

The model gives the agent its raw capability, but the harness shapes its behavior, safety, traceability, and cost. A well-designed harness can help a smaller model succeed by constraining the search space effectively. A poorly designed harness can make a capable model produce worse results than a smaller model operating with clear guardrails.

This book teaches you to build a harness, not to study models. You will not train a single model in these pages. What you will train is judgment — about where to draw boundaries, what to gate, what to observe, and what to measure.

The writing assumes you know programming and basic computer science but might not know agent-system architecture. You will build along in Rust. The runnable reference crate develops the model boundary, agent loop, tools, repository context, and policy. Later chapters widen the lens to the additional runtime concerns implemented in production *Quecto*; those chapters are architecture studies, not claims that every layer is implemented in the compact teaching crate.

This book is not documentation of one repository. It is a general guide to harness design, using Rust for implementation and *Quecto* as evidence that the architecture works.

How to Use This Book

Read Chapters 1–5 sequentially; these build the runnable reference crate. Chapters 6–12 examine production-runtime concerns, marking their exercises as design extensions when the teaching crate does not implement them. Chapter 13 brings the implemented core together and Appendix A maps the broader design to production *Quecto*.

The book uses a build-first rhythm:

1. State the system problem.
2. Define the invariant the harness must preserve.
3. Introduce the smallest workable design.
4. Add code to the shared reference implementation.
5. Demonstrate a realistic failure mode.
6. Compare the teaching design with production *Quecto*.
7. End with a focused exercise or design question.

Callouts appear throughout:

- **System invariant** — the property that must remain true.
- **Failure mode** — how a plausible implementation breaks.
- **Quecto in production** — how the real project handles the concern.
- **Beyond Rust** — how the boundary maps to other ecosystems.
- **Build checkpoint** — what the reader can run at this stage.

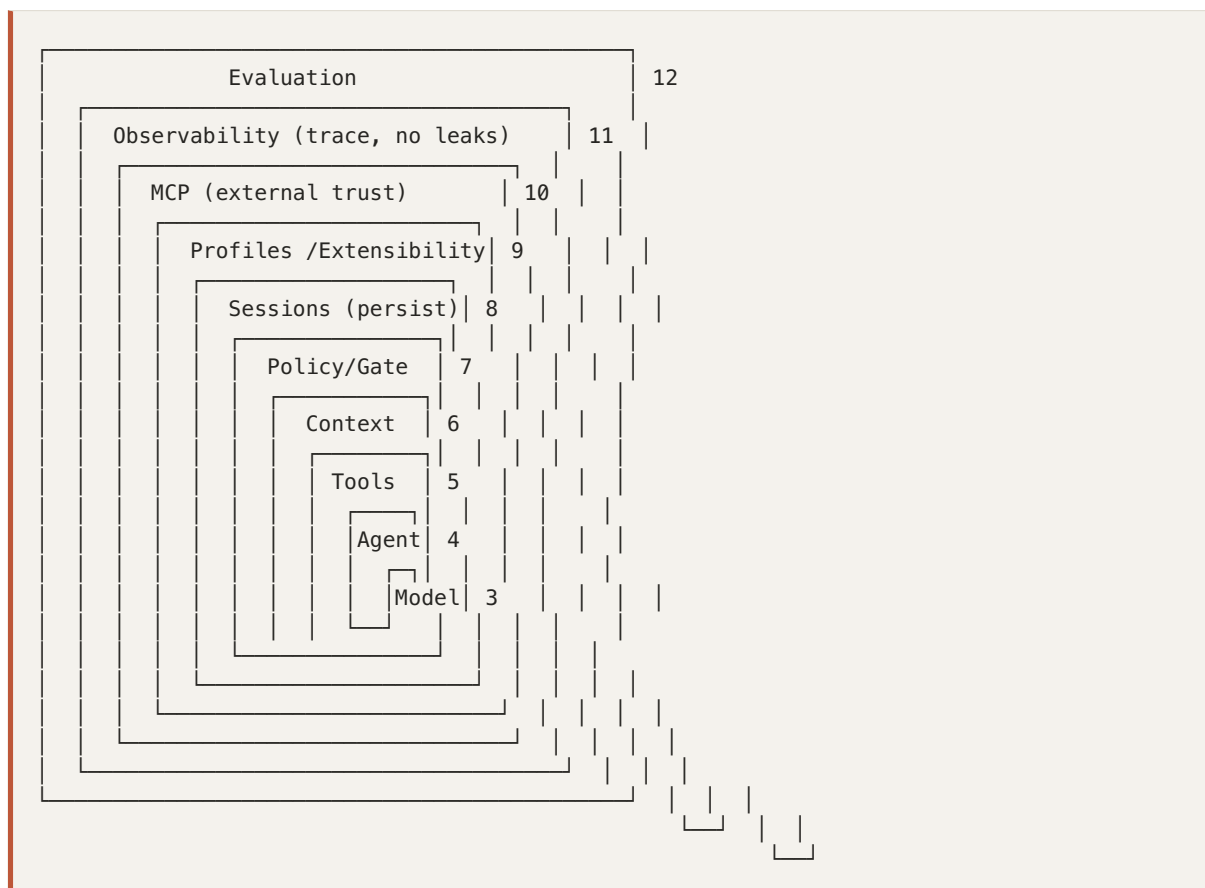
Prerequisites: Rust 2021, a Unix terminal, familiarity with HTTP and basic system programming. Python or TypeScript familiarity is helpful but not required.

Architecture Map (Summary)

The system can be understood as cooperating concerns. The first five are implemented in the teaching crate; the later runtime concerns are developed as designs and compared with Quecto:

LAYER	PURPOSE	CHAPTER
Model transport	Reach a model provider	2
Agent loop	Reason through steps safely	3
Tools	Typed communication	4
Policy	Gate reads, edits, commands	5
Verification	Confirm before finishing	6
Context	Resolve repository paths safely	7
Runtime architecture	Instructions, sessions, flavors, MCP, telemetry, evaluation	8–12
Reference harness	Assemble the runnable teaching core	13

The map below is conceptual. The runnable crate has no modules for verification retries, persisted sessions, flavors, MCP, telemetry, or evaluation.



How This Book Is Structured

The book is divided into four parts:

Part I — Build the smallest useful core (Chapters 1–3): Model transport and the agent loop.

Part II — Add tools and safety boundaries (Chapters 4–7): Typed tools, policy, verification concepts, and repository context.

Part III — Turn a loop into a runtime (Chapters 8–11): Sessions, profiles, MCP, and observability.

Part IV — Know whether it works (Chapters 12–13): Evaluation, final assembly.

Notes

- Code excerpts are drawn from the reference implementation. Long listings are in the source; short excerpts appear inline.
- The production comparison (Quecto) links to the repository. Source paths and the pinned revision are listed in Appendix A.

- "Beyond Rust" notes map interfaces to Python, TypeScript, and Go without creating parallel implementations.
-

This book is released under the MIT License. The reference implementation uses `serde`, `serde_json`, `ureq`, and `tempfile`.

1. The Harness Is the System

Chapter 1 teaches you to separate *capability* from *behavior*, to define the boundaries that determine what an agent will (and will not) do, and to establish invariants that constrain the system even when the model drifts.

The Problem

You can buy a model that writes Python, solves equations, and reasons about code. You can prompt it to "fix a bug" and it will attempt to do so. What is missing is the harness: the system that *constrains* what the model attempts, *watches* what it does, *measures* whether it succeeded, and *controls* what resources it touches.

A harness decides:

- Which tools the model may call.
- Which files the model may read or write.
- How many steps the agent may take before giving up.
- When the agent has repeated an action without progress.
- Which commands are permitted and which are blocked.
- Whether the output passes verification (tests, linters).
- How many verification attempts it gets before reporting failure.

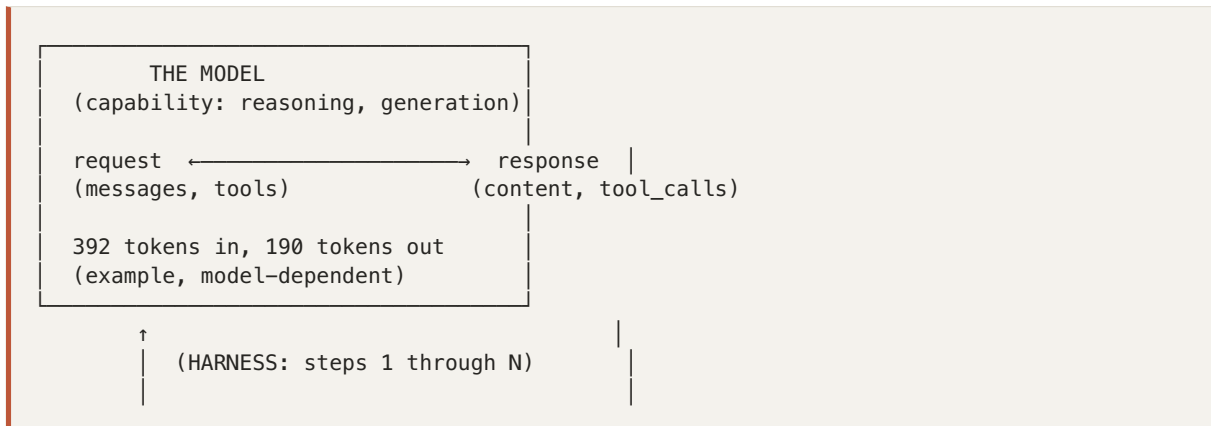
Without a harness, you have a model that may write `rm -rf /` and you have no way to stop it.

With a harness, you have a model that may *try* to write `rm -rf /` and the harness denies the call before it executes.

The harness is the system. The model is only the model.

The Model Boundary

The boundary between the model and the harness is the **API call**. This is a request (messages + tools) and a response (content + tool calls). Everything between the API call and the next API call is the harness.



The harness sits between calls. It transforms the response into a decision (what to do next), executes that decision (tools), and accumulates the result (messages) for the next model call.

System invariant: The harness must preserve the contract of the API. The response it receives from the model must be parseable as either text output or tool calls — no unhandled format is acceptable.

The Message Protocol

The model protocol is simple: a list of messages (system, user, assistant, tool), each with a role, a content string, and optional tool calls. The tool call returns a tool result. The harness builds the message list, sends it, and processes the response.

```

#[derive(Clone, Debug, PartialEq)]
pub struct Message {
    pub role: String,
    pub content: String,
    pub tool_calls: Vec<ToolCall>,
    pub tool_call_id: Option<String>,
}

impl Message {
    fn plain(role: &str, content: impl Into<String>) -> Self {
        Self {
            role: role.to_owned(),
            content: content.into(),
            tool_calls: Vec::new(),
            tool_call_id: None,
        }
    }

    pub fn system(content: impl Into<String>) -> Self {
        Self::plain("system", content)
    }

    pub fn user(content: impl Into<String>) -> Self {
        Self::plain("user", content)
    }

    pub fn assistant(content: impl Into<String>) -> Self {
        Self::plain("assistant", content)
    }

    pub fn assistant_with_calls(
        content: impl Into<String>,
        tool_calls: Vec<ToolCall>,
    ) -> Self {
        Self {
            role: "assistant".to_owned(),
            content: content.into(),
            tool_calls,
            tool_call_id: None,
        }
    }

    pub fn tool_result(tool_call_id: impl Into<String>, content: impl Into<String>) -> Self {
        Self {
            role: "tool".to_owned(),
            content: content.into(),
            tool_calls: Vec::new(),
            tool_call_id: Some(tool_call_id.into()),
        }
    }
}

#[derive(Clone, Debug, PartialEq)]
pub struct ToolCall {
    pub id: String,
    pub name: String,
    pub arguments: Value,
}

#[derive(Clone, Debug, PartialEq)]
pub struct AssistantMessage {
    pub content: String,
    pub tool_calls: Vec<ToolCall>,
}

```

```
pub finish_reason: String,
}
//
```

The `Message` type distinguishes four roles: `system`, `user`, `assistant`, and `tool`. The `ToolCall` type carries an identifier, a name, and JSON arguments. The `AssistantMessage` carries the model's response (text plus optional tool calls) and a finish reason.

Failure mode: If the harness silently drops a response field that affects control flow, it can mis-handle the turn. Decide explicitly which fields are part of the supported contract. This teaching parser validates the fields it consumes; it does not reject every unknown JSON field, which is a separate forward-compatibility choice.

The harness also sends tool definitions (JSON schemas) alongside the messages. This is how the model knows *what* tools exist and *what* arguments each accepts.

The Model Trait

The model trait is the single interface between the harness and the model:

```
pub trait Model: Send {
    fn complete(
        &mut self,
        messages: &[Message],
        tools: &[Value],
    ) -> Result<AssistantMessage, ModelError>;
}
//
```

This is intentionally narrow. The harness calls `complete` once per loop iteration and passes all messages together (full history). The model mutates its own state (session variables, token cache, stream buffer). The harness never inspects the internal state.

The `Send` bound means the model can be moved between threads if a caller chooses to do so; this teaching agent loop does not spawn a worker. The model receives tool schemas from the harness as JSON values, letting the harness control what the model sees without hard-coding tool types in the transport layer.

System invariant: The model trait must never leak memory addresses, file handles, or tokens into the message flow. The only thing crossing the boundary is messages, tool schemas, and responses.

The HTTP Transport

The default transport is a minimal HTTP client that:

- Normalizes the endpoint URL (trailing slash handling).
- Sends `Authorization: Bearer <key>` when a key is configured.
- Parses the JSON response into typed structs (`AssistantMessage` with `content`, `tool_calls`, `finish_reason`).
- Returns structured errors (`ModelError`) rather than panicking.

```

impl Model for HttpModel {
  fn complete(
    &mut self,
    messages: &[Message],
    tools: &[Value],
  ) -> Result<AssistantMessage, ModelError> {
    let mut body = Map::new();
    body.insert("model".to_owned(), Value::String(self.config.model.clone()));
    body.insert(
      "messages".to_owned(),
      Value::Array(messages.iter().map(message_to_json).collect()),
    );
    if !tools.is_empty() {
      body.insert("tools".to_owned(), Value::Array(tools.to_vec()));
    }

    let mut request = self
      .agent
      .post(&self.endpoint())
      .set("content-type", "application/json");
    if let Some(api_key) = self.config.api_key.as_deref() {
      request = request.set("authorization", &format!("Bearer {api_key}"));
    }

    let response_text = match request.send_json(Value::Object(body)) {
      Ok(response) => response
        .into_string()
        .map_err(|error| ModelError::Transport(error.to_string()))?,
      Err(ureq::Error::Status(status, response)) => {
        let body = response.into_string().unwrap_or_default();
        return Err(ModelError::HttpStatus { status, body });
      },
      Err(ureq::Error::Transport(error)) => {
        return Err(ModelError::Transport(error.to_string()));
      },
    };

    let response: Value = serde_json::from_str(&response_text)
      .map_err(|error| ModelError::InvalidResponse(error.to_string()))?;
    parse_assistant(&response)
  }
}
//
fn parse_assistant(response: &Value) -> Result<AssistantMessage, ModelError> {
  let choice = response
    .get("choices")
    .and_then(Value::as_array)
    .and_then(|choices| choices.first())
    .ok_or_else(|| ModelError::InvalidResponse("missing choices[0]".to_owned()))?;
  let message = choice
    .get("message")
    .and_then(Value::as_object)
    .ok_or_else(|| ModelError::InvalidResponse("missing choice
message".to_owned()))?;
  let content = match message.get("content") {
    None | Some(Value::Null) => String::new(),
    Some(Value::String(content)) => content.clone(),
    Some(_) => {
      return Err(ModelError::InvalidResponse(
        "message content is not a string or null".to_owned(),
      ));
    },
  }
};

```

```

let mut tool_calls = Vec::new();
if let Some(calls) = message.get("tool_calls") {
    let calls = calls.as_array().ok_or_else(|| {
        ModelError::InvalidResponse("message tool_calls is not an array".to_owned())
    })?;
    for call in calls {
        let id = call
            .get("id")
            .and_then(Value::as_str)
            .ok_or_else(|| ModelError::InvalidResponse("tool call missing id".to_owne
d()))?;
        let function = call
            .get("function")
            .and_then(Value::as_object)
            .ok_or_else(|| {
                ModelError::InvalidResponse("tool call missing function".to_owned())
            })?;
        let name = function
            .get("name")
            .and_then(Value::as_str)
            .ok_or_else(|| {
                ModelError::InvalidResponse("tool call missing function name".to_owne
d())
            })?;
        let arguments = match function.get("arguments") {
            Some(Value::String(arguments)) =>
serde_json::from_str(arguments).map_err(|error| {
                ModelError::InvalidResponse(format!(
                    "tool call arguments are not valid JSON: {error}"
                ))
            })?,
            Some(Value::Object(arguments)) => Value::Object(arguments.clone()),
            _ => {
                return Err(ModelError::InvalidResponse(
                    "tool call missing arguments".to_owned(),
                ));
            }
        };
        tool_calls.push(ToolCall {
            id: id.to_owned(),
            name: name.to_owned(),
            arguments,
        });
    }
}

Ok(AssistantMessage {
    content,
    tool_calls,
    finish_reason: choice
        .get("finish_reason")
        .and_then(Value::as_str)
        .unwrap_or_default()
        .to_owned(),
})
}
//

```

The error type distinguishes four classes:

- **InvalidConfig** — the harness configuration is wrong (missing base URL, empty model name).

- `Transport` — the network call itself failed (DNS, TLS handshake, timeout).
- `HttpStatus` — the server returned a non-200 status (401, 500, etc.).
- `InvalidResponse` — the model's response structure was unexpected (no `choices[0]`).

Failure mode: If the harness silently accepts any HTTP response (treating a 400 Bad Request as a "model is thinking" signal), the agent will loop forever on a configuration error. Every non-200 response *must* be surfaced as an error.

Quecto in Production

Quecto's production `quecto-agent` crate defines its own related `Model` interface; it is not the same Rust trait as this teaching crate. Its production path adds:

- Streaming output (incremental token-by-token rendering).
- OpenAI-compatible and Anthropic wire formats, plus configurable providers such as local OpenAI-compatible servers.
- Optional native reasoning controls and token-usage telemetry.
- Session persistence, verification, and optional OpenTelemetry integration at the agent layer.

The teaching harness omits streaming (the book's primary path is synchronous). See the "Beyond Rust" note below for how these features map to Python (using `httpx` streams) or TypeScript (using `ReadableStream`).

Exercise: Extend the Model

Write a unit test that sends a 401 Unauthorized response and verifies the harness returns `ModelError::HttpStatus`. Then modify `HttpModel::new` to reject empty `base_url`. Push these tests to the reference harness.

Beyond Rust

CONCERN	PYTHON	TYPESCRIPT	GO
<code>Model::complete</code>	<code>openai::client.complete(messages, tools)</code>	<code>openai.Client.chat.completions.create</code>	<code>openai.Cl</code>
<code>Send</code> bound	Not needed (GIL)	<code>Promise</code> in <code>async/await</code>	<code>sync</code>
Streaming	<code>response.iter_lines()</code>	<code>ReadableStream</code>	<code>http.Resp</code>

Build checkpoint

You can build the crate and run the model transport tests:

```
cd books/building-a-coding-agent-harness/examples/reference-harness
cargo test --test model_http
```

The model transport integration tests should pass. Run `cargo test` to execute the full reference-crate suite.

2. One Model Call, No Framework

Chapter 2 teaches you to treat the model as a **fidelity source of truth**, not as a co-scientist. The model gives you output; the harness gives you safety. You will build a model transport that preserves the model's raw behavior (including its failures) while surfacing errors in a structured way.

The Problem

Many agent frameworks add layers *between* the user and the model: a "safety filter" that rewrites prompts, a "pre-processor" that reformats messages, a "post-processor" that claims to improve the output. These layers are invisible to the harness writer and may contradict the harness's own invariants.

A minimal harness says: the model gives you what the model gives you. The harness decides what to *do* with it. The model is not trusted to decide what to *say* (that is the harness's job: policy, tools, constraints). The model is trusted to *generate text* (that is its job).

The boundary: **one model call per loop iteration, full history sent together.**

Configuration

The model transport is configured with a single struct:

```
#[derive(Clone, Debug)]
pub struct ModelConfig {
    pub base_url: String,
    pub api_key: Option<String>,
    pub model: String,
    pub timeout_secs: u64,
}
//
```

`ModelConfig` carries four fields:

- `base_url` — the provider endpoint (normalize to always append `/v1/chat/completions`).
- `api_key` — optional; omitting it is valid for local providers (Ollama, LM Studio).
- `model` — the model name (e.g., `"qwen3.8-35b"`).
- `timeout_secs` — how long to wait before giving up.

Failure mode: If `base_url` is empty and `timeout_secs` is `0`, the `HttpModel::new` constructor returns `ModelError::InvalidConfig`. A zero timeout means the request will *always* fail (the HTTP client will abort immediately). This is not a runtime bug; it is a configuration bug surfaced at initialization.

HTTP Round-Trip

The harness constructs the request body as a JSON object with `model` and `messages`. If tools exist, it adds a `tools` array. It sends a `POST` with `Content-Type: application/json` and an optional `Authorization` header.

```
impl Model for HttpModel {
  fn complete(
    &mut self,
    messages: &[Message],
    tools: &[Value],
  ) -> Result<AssistantMessage, ModelError> {
    let mut body = Map::new();
    body.insert("model".to_owned(), Value::String(self.config.model.clone()));
    body.insert(
      "messages".to_owned(),
      Value::Array(messages.iter().map(message_to_json).collect()),
    );
    if !tools.is_empty() {
      body.insert("tools".to_owned(), Value::Array(tools.to_vec()));
    }

    let mut request = self
      .agent
      .post(&self.endpoint())
      .set("content-type", "application/json");
    if let Some(api_key) = self.config.api_key.as_deref() {
      request = request.set("authorization", &format!("Bearer {api_key}"));
    }

    let response_text = match request.send_json(Value::Object(body)) {
      Ok(response) => response
        .into_string()
        .map_err(|error| ModelError::Transport(error.to_string()))?,
      Err(ureq::Error::Status(status, response)) => {
        let body = response.into_string().unwrap_or_default();
        return Err(ModelError::HttpStatus { status, body });
      },
      Err(ureq::Error::Transport(error)) => {
        return Err(ModelError::Transport(error.to_string()));
      }
    };

    let response: Value = serde_json::from_str(&response_text)
      .map_err(|error| ModelError::InvalidResponse(error.to_string()))?;
    parse_assistant(&response)
  }
}
//
```

The `endpoint` method normalizes the URL: if the base URL already ends with `/v1`, it appends `/chat/completions`; otherwise it appends `/v1/chat/completions`. This handles both `https://api.openai.com/v1` and `https://api.openai.com` correctly.

System invariant: The request body must be a valid OpenAI-compatible format. The harness must not add custom fields (like `temperature`, `max_tokens`) unless they are explicitly supported by the target provider. Do not assume a model supports `temperature=0.5` just because another model does.

Parsing the Response

The model returns a JSON body. The harness parses it into an `AssistantMessage`:

```

fn parse_assistant(response: &Value) -> Result<AssistantMessage, ModelError> {
    let choice = response
        .get("choices")
        .and_then(Value::as_array)
        .and_then(|choices| choices.first())
        .ok_or_else(|| ModelError::InvalidResponse("missing choices[0]".to_owned()))?;
    let message = choice
        .get("message")
        .and_then(Value::as_object)
        .ok_or_else(|| ModelError::InvalidResponse("missing choice
message".to_owned()))?;
    let content = match message.get("content") {
        None | Some(Value::Null) => String::new(),
        Some(Value::String(content)) => content.clone(),
        Some(_) => {
            return Err(ModelError::InvalidResponse(
                "message content is not a string or null".to_owned(),
            ));
        }
    };
    let mut tool_calls = Vec::new();
    if let Some(calls) = message.get("tool_calls") {
        let calls = calls.as_array().ok_or_else(|| {
            ModelError::InvalidResponse("message tool_calls is not an array".to_owned())
        })?;
        for call in calls {
            let id = call
                .get("id")
                .and_then(Value::as_str)
                .ok_or_else(|| ModelError::InvalidResponse("tool call missing id".to_owne
d()))?;
            let function = call
                .get("function")
                .and_then(Value::as_object)
                .ok_or_else(|| {
                    ModelError::InvalidResponse("tool call missing function".to_owned())
                })?;
            let name = function
                .get("name")
                .and_then(Value::as_str)
                .ok_or_else(|| {
                    ModelError::InvalidResponse("tool call missing function name".to_owne
d())
                })?;
            let arguments = match function.get("arguments") {
                Some(Value::String(arguments)) =>
serde_json::from_str(arguments).map_err(|error| {
                    ModelError::InvalidResponse(format!(
                        "tool call arguments are not valid JSON: {error}"
                    ))
                })?,
                Some(Value::Object(arguments)) => Value::Object(arguments.clone()),
                _ => {
                    return Err(ModelError::InvalidResponse(
                        "tool call missing arguments".to_owned(),
                    ));
                }
            };
            tool_calls.push(ToolCall {
                id: id.to_owned(),
                name: name.to_owned(),
                arguments,
            });
        }
    }
}

```

```

    }
  }

  Ok(AssistantMessage {
    content,
    tool_calls,
    finish_reason: choice
      .get("finish_reason")
      .and_then(Value::as_str)
      .unwrap_or_default()
      .to_owned(),
  })
}
//

```

The parser extracts:

- `content` — the text (empty string if null).
- `tool_calls` — a list of call objects (empty list if none).
- `finish_reason` — `"stop"`, `"length"`, or `"tool_calls"` (or empty string if missing).

Failure mode: If the model returns `finish_reason: "content_filter"` (content filtered out by the provider), the harness treats the text as empty. Some frameworks interpret this as a *request for user feedback*. The harness here is silent: no text, no tool calls, step ends (and the next loop iteration will send the same messages). This is intentional — we do not want the agent to *retry* a filtered prompt without understanding *why* it was filtered.

The `parse_assistant` function returns `ModelError::InvalidResponse` if the response is structurally wrong (no `choices`, no `message`, non-string content, or invalid tool call arguments). A malformed response is *never* treated as "the model gave no output."

Provider Differences

Different providers behave differently:

PROVIDER	TOOL CALL FORMAT	FINISH_REASON ON TOOL CALLS
OpenAI	<code>{"id":"call_...", "function": {"name":"read_file","arguments":"..."}}</code>	<code>"tool_calls"</code>
Anthropic	<code>{"id":"toolu_...", "input":{"path":"..."}, "type":"tool_use"}</code>	(no <code>finish_reason</code> , tool calls implied)
Ollama	Same as OpenAI (OpenAI-compatible endpoint)	<code>"tool_calls"</code>

The teaching harness uses the OpenAI format. If you target Anthropic, you must parse a slightly different structure (`input` is a raw object, not a JSON string). The `Beyond Rust` note below shows the mapping.

Quecto in production: Quecto supports provider adapters that normalize all responses to a single internal format before parsing. This isolates provider-specific parsing from the harness's core logic.

Exercise

Write a mock model that returns a response with `finish_reason: "content_filter"` and an empty content string. Verify the harness returns `Outcome::Complete("")`.

Beyond Rust

LAYER	PYTHON	TYPESCRIPT
Transport	<code>openai::client.chat.completions.create(messages=msg, tools=tools)</code>	<code>openai-js</code> or <code>fetch</code>
Config	<code>base_url</code> (default <code>"https://api.openai.com/v1"</code>)	Same, or <code>OPENAI_API_KEY</code> env var
Timeout	<code>timeout=float(options.timeout)</code>	<code>AbortSignal.timeout(timeout)</code>

Build checkpoint

```
cd books/building-a-coding-agent-harness/examples/reference-harness
cargo test --test model_http
```

Test `returns_text_reply_and_sends_openai_compatible_body` asserts the request body matches the OpenAI spec. Run it in isolation:

```
cargo test --test model_http returns_text_reply
```

3. The Bounded Agent Loop

Chapter 3 teaches you to wrap a single model call in a loop that: (1) executes tool calls the model produces, (2) accumulates results, (3) detects when the model is stuck repeating itself, and (4) terminates safely. This is the core algorithm of any coding agent: ask, act, measure, repeat — bounded.

The Problem

A model call returns a response with a list of tool calls (or just text). The harness must:

1. Execute each tool call (run a file edit, a command, a read).
2. Record the tool result.
3. Send the accumulated message list back to the model.
4. Repeat until the model says "done" (no tool calls).

Without a loop bound, the model can:

- Loop forever (repeating the same tool call).
- Execute too many tools (cost blowup, time blowup).
- Get stuck in a verification loop (retrying a failing test forever).

The loop must terminate *bounded by steps, by repetition, or by policy denial*.

The Loop

The agent is initialized with a model, a tool registry, a context (file system), and a configuration.

The `run` method receives a prompt string, pushes it as a user message, and then enters the loop:

```

pub fn run(&mut self, prompt: &str) -> Outcome {
  self.messages.push(Message::user(prompt));
  let mut repeat_guard = RepeatGuard::default();
  let mut denial_count: usize = 0;

  for _ in 0..self.config.max_steps {
    let reply = match self
      .model
      .complete(&self.messages, &self.registry.schemas())
    {
      Ok(reply) => reply,
      Err(error) => return Outcome::Error(error.to_string()),
    };
    self.messages.push(Message::assistant_with_calls(
      reply.content.clone(),
      reply.tool_calls.clone(),
    ));

    if reply.tool_calls.is_empty() {
      return Outcome::Complete(reply.content);
    }

    for call in reply.tool_calls {
      // Apply the default ReadOnly policy when denial_limit > 0.
      if self.config.denial_limit > 0 {
        let policy = Policy::from_preset(Preset::ReadOnly);
        match policy.decide(&call) {
          Decision::Allow => {}
          Decision::Ask => {
            // Waiting for user approval: do NOT execute.
            // Report the skipped action so the model protocol remains
            complete.
            self.messages.push(Message::tool_result(
              call.id.clone(),
              "policy requires user approval; the operation was not
            executed",
            ));
            // An Ask decision is not a denial and does not increment
            denial_count.
            continue;
          }
          Decision::Deny(reason) => {
            denial_count += 1;
            if denial_count >= self.config.denial_limit.max(1) {
              return Outcome::Blocked;
            }
            self.messages.push(Message::tool_result(
              call.id.clone(),
              format!("policy denied: {reason}"),
            ));
            continue;
          }
        }
      }
    }

    let result = match self.registry.execute(&call, &mut self.context) {
      Ok(output) => output.content,
      Err(error) => error.to_string(),
    };
    self.messages.push(Message::tool_result(
      call.id.clone(),
      result.clone(),
    ));
    denial_count = 0;
  }
}

```

```

        if repeat_guard.observe(
            &call,
            &result,
            self.context.changes().len(),
            self.config.repeat_limit,
        ) {
            return Outcome::RepeatedAction;
        }
    }

    Outcome::StepLimit
}
//

```

Each iteration:

1. Send all accumulated messages to the model.
2. Push the model's response (assistant message with tool calls).
3. For each tool call: execute it (via the registry), push the tool result.
4. Repeat until the model returns no tool calls (success), the step limit is reached, or the policy denies too many calls in a row.

System invariant: Every tool call in the response must correspond to a tool registered in the registry. An unknown tool name is a *tool error*, not a panic. The harness must never crash on an unknown tool name.

The Repeat Guard

The `RepeatGuard` struct detects when the model is stuck in a feedback loop: it sends the same tool call with the same arguments and receives the same output — repeated 3 or more times in a row (configurable via `repeat_limit`).

```
// Inside agent.rs (RepeatGuard struct)
```

The guard computes a *fingerprint* from (call name, call arguments, tool result). If the fingerprint matches the previous call *and* the file system has not changed (checked via `context.changes().len()`), the repetition count increments. Once the streak reaches `repeat_limit`, the agent returns `Outcome::RepeatedAction`.

Failure mode: A naive guard that only checks the call name (ignoring arguments and result) will detect distinct writes as identical. The fingerprint must include the full call + full result. The file-system-change counter is the critical differentiator: writing "file v1" and "file v2" are *different* even with the same tool name.

Termination Conditions

The agent returns one of these outcomes:

OUTCOME	WHEN
<code>Complete(String)</code>	Model returned no tool calls (text only).
<code>StepLimit</code>	Exceeded <code>max_steps</code> iterations without a text-only response.
<code>RepeatedAction</code>	Model sent the same tool call 3+ times with unchanged results.
<code>Blocked</code>	Policy denied 3+ tool calls in a row (configurable via <code>denial_limit</code>).
<code>Cancelled</code>	Reserved variant; <code>Agent::run</code> does not currently return it.
<code>VerificationFailed { attempts }</code>	Reserved variant; verification retries are not implemented in this crate.
<code>Error(String)</code>	A non-recoverable error occurred (model transport failure, file-system I/O).

System invariant: `StepLimit` is not success, but it also does not automatically roll back edits in the reference crate. The harness returns the outcome and keeps the recorded changes in context; a caller must decide how to present, inspect, or revert those changes.

Policy Gating (Optional)

When `denial_limit > 0`, each tool call is checked against a default `ReadOnly` policy. If the call is denied (e.g., the model calls `write_file` or `run_command` under a read-only preset), the harness records the denial and increments the denial counter. After `denial_limit` consecutive denials, the agent returns `Outcome::Blocked`.

This is an *optional* safety layer. It is not enabled by default (`denial_limit: 0` means no gating). In the teaching implementation, an `Ask` decision does not prompt a person; it returns an explanatory tool result and leaves the operation unexecuted. A real interactive approval flow needs a separate approver component.

Qecto in production: Qecto resolves layered TOML *flavors* (`quecto-agent/src/flavor.rs`) and combines them with approval presets and per-tool overrides. The teaching crate stops at three fixed presets and does not implement flavor loading or interactive approval.

Exercise

Write a test that:

1. Creates a scripted model returning 3 identical `echo` calls (same arguments, same result).
2. Registers the test `EchoTool` in the registry.
3. Sets `repeat_limit: 3`.
4. Verifies the agent returns `Outcome::RepeatedAction`.

You should be able to find an equivalent test in `tests/agent_loop.rs` (test name: `agent_stops_after_three_identical_tool_observations`).

Beyond Rust

CONCERN	PYTHON	TYPESCRIPT
Loop bound	<code>for _ in range(max_steps):</code>	<code>while (step < max_steps) { ... }</code>
Repeat guard	Track <code>(tool_name, args, result)</code> tuple in a list	Map
State	<code>agent.messages</code> (mutated in place)	<code>const history: Message[]</code>

Build checkpoint

```
cd books/building-a-coding-agent-harness/examples/reference-harness
cargo test --test agent_loop agent_stops_after_three_identical_tool_observations
```

This test verifies that repeated identical tool calls (no file-system change) trigger the repeat guard.

4. Typed, Unsafe-Free Communication

Chapter 4 teaches you to design a tool interface that is: (1) *typed* (the harness knows the schema before the model speaks), (2) *safe* (unregistered tools fail closed, not with a panic), and (3) *observable* (every tool produces output the harness can record).

The Problem

A raw text prompt ("delete /tmp") is unsafe: the model's intent is ambiguous. A typed tool definition ("delete_file: {path: string} → {error: null | string}") improves the boundary because:

- The harness knows *which* tool names it exposes before the model speaks.
- The tool implementation validates its own required arguments before performing I/O. A published JSON schema informs the model, but is not a substitute for validation at execution time.
- The harness can intercept or deny the call based on policy (see Chapter 5).

A tool is a callable unit of work. It has a name, a description, a JSON schema, and a run function. The harness *exposes* the tool to the model via the tool schema, and *executes* the tool when the model requests it.

The Tool Trait

The tool trait is the interface the harness uses to discover, register, and execute tools:

```
pub trait Tool: Send + Sync {
    fn name(&self) -> &'static str;
    fn description(&self) -> &'static str;
    fn schema(&self) -> Value;
    fn run(&self, args: &Value, cx: &mut Context) -> ToolResult;
}
```

Four methods:

- `name()` — the unique identifier (lowercase, underscore-separated).
- `description()` — a human-readable summary (used in the schema).
- `schema()` — the JSON Schema for arguments (used by the model).
- `run(args, context)` — the execution function (takes arguments, returns output).

The `Send + Sync` bounds make a tool eligible to be moved or shared across threads; the teaching agent loop itself does not spawn worker threads.

System invariant: A tool's `run` method must never panic. If a tool encounters an error (file not found, command timeout), it must return `ToolError::Failed(...)` — never a Rust `panic!()`. A panic inside a tool execution corrupts the agent's state and forces a restart.

The Tool Registry

The registry is a `BTreeMap` from tool name to a boxed tool trait object. It provides:

- `register(tool)` — add a tool.
- `schemas()` — generate the JSON schema list (sent to the model).
- `execute(call, context)` — find and run a tool by name.

```
impl ToolRegistry {
    pub fn new() -> Self {
        Self::default()
    }

    pub fn register(&mut self, tool: Box<dyn Tool>) {
        self.tools.insert(tool.name().to_owned(), tool);
    }

    pub fn schemas(&self) -> Vec<Value> {
        self.tools
            .values()
            .map(|tool| {
                json!({
                    "type": "function",
                    "function": {
                        "name": tool.name(),
                        "description": tool.description(),
                        "parameters": tool.schema(),
                    }
                })
            })
            .collect()
    }

    pub fn execute(&self, call: &ToolCall, cx: &mut Context) -> ToolResult {
        let tool = self
            .tools
            .get(&call.name)
            .ok_or_else(|| ToolError::Unknown(call.name.clone()))?;
        tool.run(&call.arguments, cx)
    }
}
//
```

The `BTreeMap` ensures deterministic ordering of tools in the schema (alphabetical). This matters because the model's output depends on the order of tools it sees.

Failure mode: If the registry silently ignores a duplicate registration (allowing two tools to share the same name), the last-registered tool silently wins. The harness should *log* a duplicate-name warning but *not* reject the registration (the programmer made a mistake, the harness should tolerate it).

Built-in Tools

The reference harness includes four tools:

TOOL	OPERATION	EXAMPLE
<code>ReadFile</code>	Read a file (path must be inside repo).	<code>read_file(path: "src/main.rs")</code>
<code>WriteFile</code>	Write/overwrite a file (records undo snapshots).	<code>write_file(path: "src/main.rs", content: "...")</code>
<code>ApplyPatch</code>	Replace the first occurrence of <code>old</code> text with <code>new</code> .	<code>apply_patch(path: "src/main.rs", old: "fn foo()", new: "fn bar()")</code>
<code>RunCommand</code>	Run a command through the context's timeout and output limits.	<code>run_command(command: "cargo test")</code>

Each tool records a `FileChange` (before/after content) in the context. This enables:

- Undo (revert a file to its previous state).
- Diff (see what the agent changed in a single step).
- Verification (diff the output of the agent's edits against a baseline).

Quecto in production: Quecto adds `SearchText`, `ListFiles`, `GitDiff`, and `GitStatus` around the file, patch, and shell tools. The exported tool surface is listed in `quecto-agent/src/lib.rs` and implemented under `quecto-agent/src/tools/`.

Quecto's Version

Quecto's tools extend this interface with repository search, file listing, Git status/diff inspection, shell execution, and optional MCP-backed tools.

See the "Beyond Rust" note for how Quecto's tool system maps to Python (using `openai::tools` parameter with a custom tool schema).

Exercise

Write a tool `AppendFile` that appends text to the end of a file (creating the file if it does not exist). Register it in the test suite and verify that:

1. Reading the file before writing returns the original content.
2. Reading the file after writing returns the appended content.
3. The context records a `FileChange` with the before/after.

You can find an equivalent pattern in `tests/safety.rs` (test name: `write_file_is_atomic_and_records_ordered_undo_snapshots`).

Beyond Rust

CONCEPT	PYTHON	TYPESCRIPT
<code>Tool::schema()</code>	<code>{"type":"object","properties":{...}}</code>	Same
<code>Tool::run()</code>	Custom callable object	Custom class/method
Registry	<code>dict[str, Tool]</code>	<code>Map<string, Tool></code>

Build checkpoint

```
cd books/building-a-coding-agent-harness/examples/reference-harness
cargo test --test agent_loop registry_emits_schemas_in_name_order
```

This test verifies that the registry sorts tool schemas alphabetically (`echo` < `zeta`).

5. Execution Policy and Approval Gates

Chapter 5 teaches you to design a *gate* that decides *whether* a tool call should execute. This gate is separate from the tool itself (a tool's job is to *do* something; the policy's job is to *decide* if it may). The gate supports three presets: readonly, editor, and full — each with increasing permissions.

The Problem

A model will happily call `rm -rf /` if it thinks the file path matches. A model will also call `git push origin main` when you asked it to "fix a bug" (it thought you wanted to push the fix). Without a policy, the harness executes every tool call the model makes.

The policy layer answers: *should this tool call execute?*

The policy returns one of three decisions:

- **Allow** — execute immediately.
- **Ask** — require an external approval decision before execution. The compact teaching agent has no interactive approver; it records an explanatory tool result and does not execute the call.
- **Deny** — reject unconditionally (the tool is unknown, or the command is hard-blocked).

System invariant: The policy must be a *pure function* of the call and the preset. It must not inspect the file system (no file I/O in the policy decision). If it does, a file system error could *throttle* the agent (a file-system "file not found" error causes the policy to reject all calls).

The Decision Enum

```
pub enum Decision {
    Allow,
    Ask,
    Deny(String),
}
```

- `Allow` — proceed without interruption.
- `Ask` — the user has authorized this operation but the preset does not permit it directly. The harness displays a prompt: "The agent wants to call ``write_file("main.rs", ...)`. Approve?" The user (or a stored preset) responds Yes/No.
- `Deny(String)` — the call is unconditionally rejected. The string explains why (used in logs and in the agent's "Blocked" message).

Presets

Three presets encode safety trade-offs:

PRESET	READFILE	WRITEFILE	APPLYPATCH	RUNCOMMAND
<code>ReadOnly</code>	✓ (Allow)	✗ (Ask)	✗ (Ask)	✗ (Ask)
<code>Editor</code>	✓ (Allow)	✓ (Allow)	✓ (Allow)	✗ (Ask)
<code>Full</code>	✓ (Allow)	✓ (Allow)	✓ (Allow)	✓ (Allow)*

**Full does not mean all commands are allowed. It means all tools are allowed. The hard-denied command list (below) still applies.*

Quecto in production: Quecto resolves layered TOML *flavors* (`quecto-agent/src/flavor.rs`) and has approval presets and per-tool overrides. The teaching crate stops at three fixed presets and hard-deny patterns; it does not implement flavor loading or interactive approval.

Hard-Denied Commands

Even under the `Full` preset, certain commands are always blocked:

PATTERN	EXAMPLE
<code>sudo</code> (with space)	<code>sudo rm -rf /</code>
<code>sudo\$</code>	<code>sudo; rm -rf /</code>
<code>git push</code> (with space)	<code>git push origin main</code>
<code>sh -c</code>	<code>sh -c 'rm -rf /'</code>

These are patterns, not exact matches. A command containing any of these substrings is rejected. This is a conservative heuristic: it may *over-block* (e.g., a legitimate `sudo docker run --rm ...` is blocked because it contains "sudo "), but it avoids *under-blocking* (a destructive command slips through).

Failure mode: If the pattern matching is too loose (matching "prefix" substring in "file-prefixed.log"), the agent cannot run legitimate commands. The heuristic should be tuned empirically. A better approach is *token-based* matching (match the first token against a block list) rather than substring matching.

The Decide Method

The `Policy::decide(call)` method follows a simple three-step logic:

1. **Check the hard-denied list.** If the command matches a pattern, return `Deny("hard-denied command")`. (The check is performed on the `command` argument of the call.)
2. **Is the tool known?** A tool is "known" if it appears in *any* preset (across all presets). If the tool name is not known, return `Deny("unknown tool 'name'")`.
3. **Is the tool in the current preset?** If yes, `Allow`. If no (it is known but not in this preset), `Ask`.

```

use crate::model::ToolCall;
use serde_json::Value;
use std::collections::BTreeSet;

/// Outcome of a policy gate.

#[derive(Clone, Debug, PartialEq)]
pub enum Decision {
    Allow,
    Ask,
    Deny(String),
}

/// Policy presets that trade convenience against safety.

#[derive(Clone, Copy, Debug, PartialEq)]
pub enum Preset {
    ReadOnly,
    Editor,
    Full,
}

/// Hardcoded commands that even Full never permits.

const HARD_DENY: &[&str] = &[
    "sudo ", "sudo;", "sudo$",
    "git push", "git push ", "git push;",
    "sh -c", "sh -c'", "sh -c\\\"",
    "rm -rf /",
];

/// Union of ALL known tool names (used to distinguish "unknown" from "allowed in another
/// preset").

const ALL_TOOLS: &[&str] = &["read_file", "write_file", "apply_patch", "run_command"];

fn is_known_tool(name: &str) -> bool {
    ALL_TOOLS.contains(&name)
}

/// Summary of a spawned command execution.

#[derive(Clone, Debug, PartialEq)]
pub struct ToolSummary {
    pub stdout: Option<String>,
    pub stderr: Option<String>,
    pub exit_code: Option<i32>,
    pub truncated: bool,
    pub timed_out: bool,
    pub cancelled: bool,
}

/// A policy evaluator that gates tool calls according to a preset.

#[derive(Clone, Debug)]
pub struct Policy {
    presets: BTreeSet<&'static str>,
    deny_list: &'static [&'static str],
}

impl Policy {
    pub fn from_preset(preset: Preset) -> Self {
        let presets = match preset {
            Preset::ReadOnly => {

```

```

        let mut p = BTreeSet::new();
        p.insert("read_file");
        p
    }
    Preset::Editor => {
        let mut p = BTreeSet::new();
        p.insert("read_file");
        p.insert("write_file");
        p.insert("apply_patch");
        p
    }
    Preset::Full => {
        let mut p = BTreeSet::new();
        p.insert("read_file");
        p.insert("write_file");
        p.insert("apply_patch");
        p.insert("run_command");
        p
    }
};
Self {
    presets,
    deny_list: HARD_DENY,
}
}

/// Return the decision for a single tool call under this policy.

pub fn decide(&self, call: &ToolCall) -> Decision {
    // 1. Hard-denied commands always fail.
    if self.deny_list.iter().any(|d| {
        call.arguments
            .get("command")
            .and_then(Value::as_str)
            .map(|c| c.contains(d))
            .unwrap_or(false)
    }) {
        return Decision::Deny("hard-denied command".to_owned());
    }

    let known = is_known_tool(call.name.as_str());
    let in_preset = self.presets.contains(call.name.as_str());

    match (known, in_preset) {
        (true, true) => Decision::Allow,
        (true, false) => Decision::Ask,
        (false, _) => Decision::Deny(format!("unknown tool '{}'", call.name)),
    }
}
}

```

System invariant: The `ALL_TOOLS` array (union of all preset tools) must be kept in sync with the actual tools registered in the harness. If you add a new tool, add its name to `ALL_TOOLS`. Otherwise the policy will incorrectly classify it as "unknown" (and deny it).

Quecto's Version

Quecto combines the policy decision with flavor configuration, an approval mode, a terminal approver, and trust checks. These are separate production components rather than fields in the teaching `Policy` type.

The teaching harness is simpler: one policy, one preset, no overrides.

Exercise

Write a test that:

- 1. Creates a `ReadOnly` policy.
- 2. Verifies that `read_file("a")` returns `Allow`, `write_file("a", "...")` returns `Ask`.
- 3. Verifies that `run_command("printf safe")` returns `Ask`.
- 4. Creates a `Full` policy and verifies `run_command("echo ok")` returns `Allow`.

You can find equivalent tests in `tests/safety.rs` (test name: `policy_presets_gate_reads_edits_and_commands`).

Beyond Rust

CONCEPT	PYTHON	TYPESCRIPT
Decision	<code>enum: {Allow, Ask, Deny}</code>	<code>enum: { Allow, Ask, Deny }</code>
<code>Policy::decide</code>	Callable class: <code>def decide(self, call: ToolCall) -> Decision:</code>	<code>decide(call: ToolCall): Decision</code>
Preset	Dictionary mapping tool name → boolean	Record type

Build checkpoint

```
cd books/building-a-coding-agent-harness/examples/reference-harness
cargo test --test safety policy_presets_gate_reads_edits_and_commands
```

This test verifies that `ReadOnly` allows reads, asks writes, and asks commands.

6. Verification as a Completion Gate

This chapter designs a verification gate. The compact reference crate does not include verification retries or automatic rollback; Quecto's production verifier is a separate, simpler command gate.

The Problem

A model will claim its work is finished when:

- The last edit was syntactically valid.
- The model's *internal reasoning* says "I'm done."
- The model *prefers* to stop (lower token count, "stop" finish reason).

None of these are sufficient. The model may have introduced a regression that the tests will catch. The model may have deleted a dependency that a linter will flag.

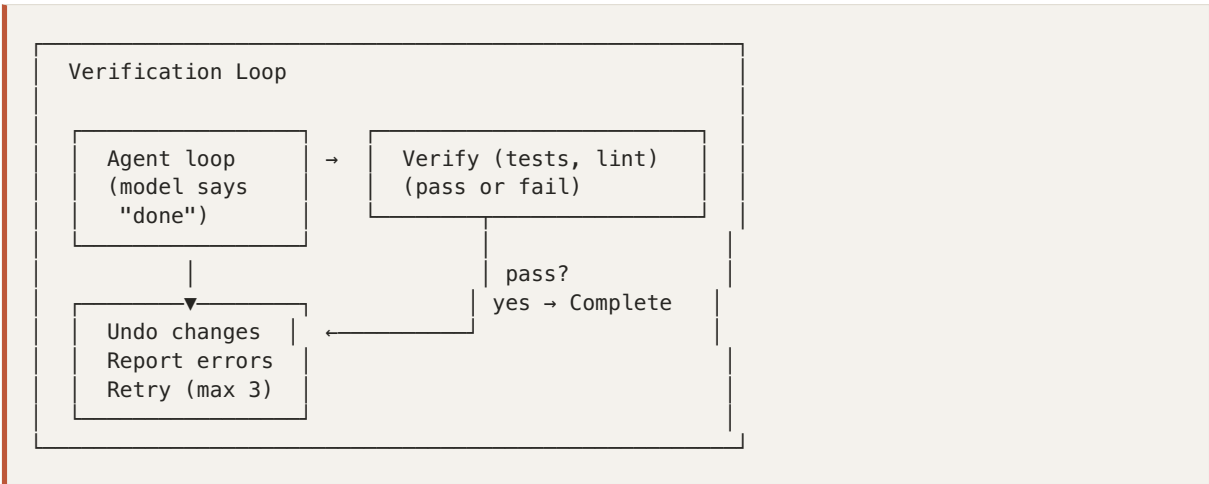
Verification is one answer: before the agent reports "done," run appropriate verification commands. If verification fails, report the result and define explicitly whether the agent may make a bounded repair attempt. Rollback is a separate policy decision; automatically discarding all edits can destroy useful work and is not a behavior of the compact teaching crate.

System invariant: The agent must never report `Outcome::Complete(...)` if verification has not been run. Verification must always run (unless explicitly disabled).

The Verification Loop

When the agent's model says "done" (no tool calls), the harness:

1. Collects all file changes since the prompt.
2. Runs the verification commands (tests, linters, etc.).
3. If verification passes → `Outcome::Complete(...)`.
4. If verification fails → report failures and, if configured, allow a bounded repair attempt.
5. If retry budget exhausted → `Outcome::VerificationFailed { attempts }`.



The Verification Command

A verification command is any command that:

- Returns exit code `0` on success.
- Returns non-zero on failure.
- Produces output that the harness can parse (test names, file paths).

Common verification commands:

- `cargo test` (Rust unit tests).
- `cargo clippy` (Rust linter).
- `cargo fmt --check` (formatting check).
- `npm test` (JavaScript/TypeScript tests).
- `flake8` (Python linter).
- `tsc --noEmit` (TypeScript type check).

Failure mode: If verification fails with a *structural* error (e.g, the test runner crashed with a syntax error), the harness must *not* blindly retry (the agent will undo and re-apply the same broken code). Instead, the harness should *short-circuit* and report the structural error.

Retry Budget

Each verification failure consumes one *retry*. The budget (configurable, default 3) limits how many times the agent retried after verification failures.

OUTCOME	MEANING
<code>Complete(String)</code>	Model said done, verification passed.

OUTCOME	MEANING
<code>VerificationFailed { attempts }</code>	Model said done, verification failed, retried 3 times and failed again.
(loop continues)	Model said done, verification failed, retry budget not exhausted.

Quecto in production: `quecto-agent/src/verify.rs` runs configured verification commands through the repository command boundary and reports each command's status and output. Quecto does not implement the incremental, parallel, or statistical policies sketched in earlier drafts of this chapter. Verification is a completion gate, not proof that a change is correct; choose commands that exercise the behavior you care about.

Example

Suppose the model edits `src/main.rs` and runs `cargo test`. The test output is:

```
test src/main.rs ... FAILED (assertion: x == y, got 5, expected 10)
test src/util.rs ... PASSED
```

The harness:

1. Detects one failure in `src/main.rs`.
2. *Does* call `cargo test` again (retry 1).
3. Result: same failure. Retry 2. Same failure. Retry 3. Same failure.
4. Budget exhausted → `Outcome::VerificationFailed { attempts: 3 }`.
5. The harness reports: "Verification failed 3 times: 1 test in `src/main.rs`."

The model (prompted with the failures) may then make a corrective edit (change `x = 5` to `x = 10` in `src/main.rs`).

System invariant: A retry policy must define what happens to the current diff. Do not claim that a retry implies rollback: preserve, checkpoint, or revert changes intentionally, and make the user-visible behavior explicit.

Exercise

Write a verification test that:

1. Creates a file `test.rs` with `assert_eq!(1, 2)` (always fails).
2. Runs the verification command `cargo test` (which will include `test.rs`).
3. Verifies that the harness returns `Outcome::VerificationFailed { attempts: 3 }`.

This test is a design exercise: the compact teaching crate does not yet include a `Verifier` module. To experiment, build the command-runner boundary first and keep subprocess tests opt-in where the environment restricts process creation.

Beyond Rust

VERIFICATION	PYTHON	TYPESCRIPT
Unit tests	<code>pytest</code>	<code>jest</code> , <code>mocha</code>
Linters	<code>flake8</code> , <code>pylint</code>	<code>eslint</code>
Type check	N/A (dynamic)	<code>tsc --noEmit</code>
Formatter	<code>black --check</code>	<code>prettier --check</code>

Build checkpoint

The current teaching-crate suite includes non-spawning model, tool, context, policy, and agent tests. Command execution and verification require child processes and should be run only in an environment that permits them.

7. Context and Instruction Loading

Chapter 7 teaches you to load repository instructions, system prompts, seeded context, and precedence rules *before* the agent starts. You will also learn to protect against accidental instruction mixing (e.g., a `.env` file's instructions are not the *project* instructions).

The Problem

A coding agent needs *context* before it starts working. It needs to know:

- What is the repository root?
- What are the rules (encoded in a `.rules.md` file)?
- What is the project structure (file tree)?
- What are the APIs used in this codebase?
- What commands are available (lint, test, build)?

If the agent cannot find these instructions, it must *guess*, and guessing is bad. A well-designed harness *loads context* from a standardized location (`.claude/rules.md`, `.gpt_prompt.md`, `README.md`) *before* any model call.

System invariant: The agent must *never* inject instructions that are not authorized by the current project. If the user's `.env` file contains an instruction like "always prepend the UUID header," the agent must *not* load that instruction.

The Context Object

The `Context` struct holds:

- `repo_root` — the repository root (canonicalized).
- `changes` — a list of all file modifications (before/after).
- `limits` — optional command execution limits (timeout, output size).
- `cancel_token` — a handle that allows the harness to cancel a running command.

```

use std::error::Error;
use std::fmt;
use std::io::Read as _;
use std::path::PathBuf;
use std::process::Command;
use std::sync::atomic::{AtomicBool, Ordering};
use std::sync::Arc;
use std::time::{Duration, Instant};

#[derive(Clone, Debug, PartialEq)]
pub struct FileChange {
    pub path: PathBuf,
    pub before: Option<Vec<u8>>,
    pub after: Option<Vec<u8>>,
}

#[derive(Debug)]
pub struct ContextError(pub String);

impl fmt::Display for ContextError {
    fn fmt(&self, formatter: &mut fmt::Formatter<'_>) -> Result<(), fmt::Error> {
        write!(formatter, "repository context error: {}", self.0)
    }
}

impl Error for ContextError {}

/// Token that allows the harness to cancel a running command.

#[derive(Clone, Debug)]
pub struct CancelToken {
    cancelled: Arc<AtomicBool>,
}

impl CancelToken {
    pub fn new() -> Self {
        Self {
            cancelled: Arc::new(AtomicBool::new(false)),
        }
    }

    pub fn cancel(&self) {
        self.cancelled.store(true, Ordering::SeqCst);
    }

    pub fn is_cancelled(&self) -> bool {
        self.cancelled.load(Ordering::SeqCst)
    }
}

/// Limits for a spawned subprocess.

#[derive(Clone, Debug, PartialEq)]
pub struct CommandLimits {
    pub timeout: Duration,
    pub max_output_bytes: usize,
}

/// Default limits used when the harness is created without explicit constraints.

const DEFAULT_TIMEOUT: Duration = Duration::from_secs(30);
const DEFAULT_MAX_OUTPUT: usize = 64 * 1024;

pub struct Context {

```

```

repo_root: PathBuf,
changes: Vec<FileChange>,
limits: Option<CommandLimits>,
cancel_token: Option<CancelToken>,
}

impl Context {
    pub fn new(repo_root: PathBuf) -> Result<Self, ContextError> {
        let repo_root = repo_root
            .canonicalize()
            .map_err(|error| ContextError(error.to_string()))?;
        if !repo_root.is_dir() {
            return Err(ContextError(format!(
                "{} is not a directory",
                repo_root.display()
            )));
        }
        Ok(Self {
            repo_root,
            changes: Vec::new(),
            limits: Some(CommandLimits {
                timeout: DEFAULT_TIMEOUT,
                max_output_bytes: DEFAULT_MAX_OUTPUT,
            }),
            cancel_token: Some(CancelToken::new()),
        })
    }

    pub fn with_command_limits(
        repo_root: PathBuf,
        limits: CommandLimits,
        cancel_token: CancelToken,
    ) -> Result<Self, ContextError> {
        let repo_root = repo_root
            .canonicalize()
            .map_err(|error| ContextError(error.to_string()))?;
        if !repo_root.is_dir() {
            return Err(ContextError(format!(
                "{} is not a directory",
                repo_root.display()
            )));
        }
        Ok(Self {
            repo_root,
            changes: Vec::new(),
            limits: Some(limits),
            cancel_token: Some(cancel_token),
        })
    }

    pub fn changes(&self) -> &[FileChange] {
        &self.changes
    }

    pub fn record_change(&mut self, change: FileChange) {
        self.changes.push(change);
    }

    /// Resolve a path relative to the repo root and reject anything that escapes.

    pub fn resolve_existing(&self, relative: &str) -> Result<PathBuf, ContextError> {
        let candidate = self.repo_root.join(relative);
        let canonical = match candidate.canonicalize() {
            Ok(c) => c,

```

```

        Err(_) => return Err(ContextError("file does not exist".into())),
    };
    if !canonical.starts_with(&self.repo_root) {
        return Err(ContextError("path escapes repo root".into()));
    }
    Ok(canonical)
}

/// Resolve a path for creation: the parent directory must already exist
/// inside the repo. Does not require the target file itself to exist.

pub fn resolve_for_create(&self, relative: &str) -> Result<PathBuf, ContextError> {
    let candidate = self.repo_root.join(relative);
    // If the path is a bare filename (e.g. "note.txt"), the repo root
    // itself is the parent, so skip canonicalizing the parent.
    if candidate.parent().map(|p| p.as_os_str().is_empty()).unwrap_or(false) {
        return Ok(candidate);
    }
    let parent = candidate.parent().unwrap();
    let parent_canonical = parent.canonicalize().map_err(|_| {
        ContextError(format!("parent directory does not exist: {}", parent.display()))
    })?;
    if !parent_canonical.starts_with(&self.repo_root) {
        return Err(ContextError("parent path escapes repo root".into()));
    }
    Ok(candidate)
}

/// Run a shell command subject to timeout, output truncation, and cancellation.
///
/// For a synchronous harness (like this one), the implementation uses a
/// blocking read. Short timeouts (e.g. 50 ms) only work when the harness
/// is layered with an async runtime (see the async section of the book).

pub fn run_command(&self, command: &str) -> Result<super::policy::ToolSummary, ContextError> {
    let limits = self.limits.as_ref().ok_or_else(|| {
        ContextError("command execution not configured".into())
    })?;

    let token = self.cancel_token.as_ref().ok_or_else(|| {
        ContextError("command execution not configured".into())
    })?;

    // Fast path: already cancelled.
    if token.is_cancelled() {
        return Ok(super::policy::ToolSummary {
            stdout: Some(String::new()),
            stderr: None,
            exit_code: None,
            truncated: false,
            timed_out: false,
            cancelled: true,
        });
    }

    let start = Instant::now();

    let mut cmd = Command::new("/bin/sh");
    cmd.arg("-c").arg(command);
    cmd.stdout(std::process::Stdio::piped());

    let mut handle = cmd.spawn().map_err(|e| ContextError(format!("spawn failed:

```

```

{e}")))?;

    let mut stdout = handle.stdout.take().ok_or_else(|| {
        ContextError("failed to take stdout".into())
    })?;

    let mut data = Vec::new();
    let mut buf = vec![0u8; 256];

    loop {
        // Check timeout and cancellation before reading.
        if start.elapsed() >= limits.timeout {
            handle.kill().ok();
            // Do NOT wait – the process may linger as a zombie in some
            // environments; returning early avoids a hang.
            return Ok(super::policy::ToolSummary {
                stdout: Some(String::from_utf8_lossy(&data).to_string()),
                stderr: None,
                exit_code: None,
                truncated: data.len() >= limits.max_output_bytes,
                timed_out: true,
                cancelled: false,
            });
        }
        if token.is_cancelled() {
            handle.kill().ok();
            return Ok(super::policy::ToolSummary {
                stdout: Some(String::from_utf8_lossy(&data).to_string()),
                stderr: None,
                exit_code: None,
                truncated: false,
                timed_out: false,
                cancelled: true,
            });
        }

        match stdout.read(&mut buf) {
            Ok(n) => {
                if n > 0 {
                    data.extend_from_slice(&buf[..n]);
                    if data.len() >= limits.max_output_bytes {
                        data.truncate(limits.max_output_bytes);
                    }
                } else if n == 0 {
                    break;
                }
            }
            Err(_) => break,
        }
    }

    let status = handle.try_wait().ok().and_then(|s| s.unwrap().code()).unwrap_or(-1)
;

    Ok(super::policy::ToolSummary {
        stdout: Some(String::from_utf8_lossy(&data).to_string()),
        stderr: None,
        exit_code: Some(status),
        truncated: false,
        timed_out: false,
        cancelled: false,
    })
}
}

```

The `Context::new(root)` constructor canonicalizes the root and returns an error if it is not a valid directory. The `Context::with_command_limits(root, limits, token)` constructor adds command execution support (see the `run_command` section below).

System invariant: `Context::new` must *never* panic or accept an invalid path. It must return `ContextError` (or `None` in the result type) for any invalid path.

Path Resolution

The harness exposes two path resolution methods:

METHOD	PURPOSE	BEHAVIOR
<code>resolve_existing(relative)</code>	Read a file.	Must exist (canonicalized). Rejects path escapes (canonical path outside repo).
<code>resolve_for_create(relative)</code>	Write a file.	Parent directory must exist (but the file itself need not).

System invariant: Both methods must *reject* paths that escape the repository root. If `relative = "../etc/passwd"`, the canonicalized path is outside the repo, and the method returns `ContextError`.

A path escape is any `..` component or symlink that, when resolved, points outside the repo root.

The Command Boundary

When command execution is configured (`with_command_limits`), the context exposes `run_command(command)`:

```
// Part of context.rs (full implementation)
```

The command is executed as `/bin/sh -c "command"`. The harness supports:

- **Timeout** — the command aborts after the configured duration (default: 30 seconds).
- **Output truncation** — if the output exceeds `max_output_bytes` (default: 64 KB), the output is truncated and the `truncated` flag is set.
- **Cancellation** — the `CancelToken` allows the harness to cancel a running command at any point (used in the agent's cancellation flow).

Failure mode: If the harness does not apply a timeout, a hanging `sleep 10000` command will block the agent indefinitely. Always set a timeout (default 30 seconds).

Seed Instructions

The harness can be seeded with instructions:

- **System prompt** (hard-coded in the harness).
- **Repository instructions** (from a `.claude/rules.md` file in the repo root).
- **Seeded context** (explicitly provided in the model config).

The precedence is:

1. Hard-coded system prompt (always present).
2. Repository instructions (if `.claude/rules.md` exists).
3. Seeded context (explicit, highest priority).

Quecto in production: `quecto-agent/src/instructions.rs` loads `AGENTS.md`, `CLAUDE.md`, and `.agent/instructions.md` from the repository root down to the current working directory. Root instructions are emitted first and nearer instructions later. `quecto-agent/src/context.rs` separately seeds task context; see the pinned source in Appendix A.

Exercise

Write a test that:

1. Creates a temp directory with a file `instructions.md` containing `always prepend "PRIORITY: " to output`.`
2. Verifies that the context's `resolve_existing("instructions.md")` succeeds, but the *instructions file is NOT injected* as system prompt (system prompt must come from the config, not from arbitrary files).

This test demonstrates the *separation of concerns*: instructions are input to the *user's prompt*, not the *model's system prompt*.

Beyond Rust

CONCERN	PYTHON	TYPESCRIPT
Repo root	<code>Path(os.getenv("REPO_ROOT"))</code>	<code>process.env.REPO_ROOT</code>
Instructions	<code>open(rules_path)</code>	<code>readFileSync(rules_path, 'utf-8')</code>
Seed	<code>config.seed: "..."</code>	<code>seed: "..."</code>

Build checkpoint

```
cd books/building-a-coding-agent-harness/examples/reference-harness
cargo test --test safety context_rejects_parent_and_symlink_path_escapes
```

This test verifies that the context rejects path escapes (including symlinks).

8. Sessions and State Management

This chapter designs session persistence and recovery. The `Session` APIs below are illustrative; the compact reference crate does not define this type. Quecto production uses a SQLite-backed store and a separate agent/run lifecycle.

The Problem

A raw model API is a single pair: *you* send text, it returns text. A coding agent has *state* between turns: file edits, command output, lint results, the last verification result, the retry count. If this state is fragmented across variables, the harness has *no single source of truth* about what happened — and every new feature adds another variable, doubling the chance of inconsistency.

A session is the *single mutable handle* for one coding task. It owns the message history, the file diff, the verification state, and the cancellation token. One object.

The Session Struct

Implementation boundary: This chapter presents a session design, not code from the compact reference crate. The teaching `Agent` keeps its messages in memory for one instance; production Quecto persists resumable sessions in SQLite.

A session holds:

- `id` — a UUID (for logging, for the user).
- `root` — the repository root (canonicalized).
- `context` — the file-system handle (read/write with undo snapshots).
- `messages` — the accumulated message list (user prompt, assistant tool calls, tool results).
- `changes` — the list of file changes (before/after), used by the verification gate.
- `result` — the final outcome (Complete, VerificationFailed, StepLimit, ...).

The session is created by the harness and handed to the agent. The agent *mutates* the session's message list and the context's file state. The session is *not* shared with the model — the harness reads the messages and passes them to the model, then writes the result *back* into the session.

System invariant: The session's message list must contain *every* tool call and result, in order. If a result is dropped, the model's next turn is wrong (it receives an incomplete conversation).

The Session API

METHOD	RETURNS
<code>Session::new(root)</code>	A session with an empty message list.
<code>Session::push_user(prompt)</code>	Pushes a user message (returns <code>&mut Self</code>).
<code>Session::push_assistant(messages)</code>	Pushes assistant tool calls (returns <code>&mut Self</code>).
<code>Session::push_tool_result(name, result)</code>	Pushes a tool result (returns <code>&mut Self</code>).
<code>Session::messages()</code>	Returns the full message list (for the model).
<code>Session::changes()</code>	Returns the diff (for verification).
<code>Session::result()</code>	Returns the final outcome.
<code>Session::cancel()</code>	Cancels the session (sets the cancellation token).

System invariant: `Session::push_user` must *reject* a prompt that is empty or contains only whitespace. The harness must never send an empty prompt to the model (the model's behavior on empty text is undefined and may crash).

The Execution Flow

The harness runs the session in a single loop:

```
session.push_user("fix the bug in src/main.rs")
loop:
    response = model.complete(session.messages())
    for call in response.tool_calls:
        result = context.run(call)
        session.push_tool_result(call.name, result)
    session.push_assistant(response.tool_calls)
    if response.no_tool_calls: break (model said "done")
session.result = Outcome::Complete(...) (or fail)
```

The session's `result` field is the *only* place the final outcome is stored. The harness reads `session.result()` at the end.

Failure mode: If the harness does not call `push_tool_result` (and instead discards the result into a log), the session's message list is incomplete and the model's *next* turn will be confused (it will re-send the same tool call, because it never received the answer).

The Session as a Scope

The session's lifetime is the scope of one coding task. In a production runtime, a user submits a task and the harness creates or resumes persisted state. A second task should begin with isolated state rather than accidentally inheriting another task's messages or changes.

This is the key insight: *a session is not a conversation across tasks*. It is a single task. If a user says "first fix the auth bug, then refactor the API," the harness creates *two sessions* (one after the other), not one session with 400 messages.

System invariant: A new session must not inherit another task's messages or tool-call state. Reusing a session is explicit resume; a fresh task receives a fresh identity and state boundary.

Exercise

Write a test that:

1. Creates a session with `root = temp_dir()`.
2. Writes a file `a.rs` with `fn main() {}`.
3. Pushes a user prompt "rename fn main to fn entry".
4. Calls the model (mocked to return one `WriteFile` call renaming `main` → `entry`).
5. Verifies that `session.changes()` returns exactly one change (old: `fn main() {}`, new: `fn entry() {}`).
6. Verifies that `session.result()` is `Outcome::Complete`.

This is a design exercise for a future session module. The compact crate currently tests message accumulation and tool results within the `Agent`; it does not expose a `Session` type or persist changes.

Beyond Rust

CONCERN	PYTHON	TYPESCRIPT
Session creation	<code>Session.new(root)</code>	<code>new Session(root)</code>
Message push	<code>session.push(...)</code>	<code>session.push(message)</code>
<code>no_tool_calls</code> check	<code>if not response.tool_calls: break</code>	<code>if (response.toolCalls.length === 0) break</code>

Build checkpoint

```
cd books/building-a-coding-agent-harness/examples/reference-harness  
cargo test --test agent_loop session_populates_messages_and_records_change
```

Run `cargo test` for the implemented teaching-crate checkpoints. The session exercise above is not an existing test.

9. Profiles and Security Gates

Chapter 9 designs layered per-project policy configuration. The generic "profile" model below is illustrative; the compact crate has only fixed presets, while production Quecto calls its TOML configuration *flavors*.

The Problem

The three presets (ReadOnly, Editor, Full) from Chapter 5 are a *starting point* for teaching. In production, every repository has a *different threat model*. A Rust web server is not the same security posture as a personal finance script. A "Full" policy that allows `sudo docker run` on one repo is reckless on another.

A *profile* in the generic design is a named bundle of rules: allowed tools, blocked commands, approval requirements. The policy checks every proposed tool call against the resolved configuration before execution.

System invariant: The harness must *never* execute a tool call that violates the active profile's rules. The policy decision is *not advisory* — a denied call is rejected, no approval prompt is shown, and the tool is *not executed*.

The Profile Struct

A profile is a *deterministic configuration* (no file I/O during policy decisions). It contains:

- `allowed_tools` — the set of tool names permitted (empty = no tools, `*` = all).
- `blocked_commands` — patterns of shell commands that are always blocked (e.g., `sudo`, `git push`).
- `approval_required` — a set of (tool, path) pairs that require human approval, even if the tool is allowed.
- `profiles` — a map of *sub-profiles* (named overrides for specific tools).

Implementation boundary: Profiles are a design chapter, not a module in the compact teaching crate. In Quecto, the corresponding production feature is called a *flavor* and is implemented in `quecto-agent/src/flavor.rs`.

In a profile-based design, load the active configuration before the run rather than allowing it to change unpredictably between tool calls. Quecto uses layered TOML flavor configuration; the compact reference crate has no configuration-file loader.

Failure mode: If configuration changes while a run is in progress, the same tool call may be allowed under one policy state and denied under another. Snapshot the resolved policy at a clearly defined boundary and surface parse errors rather than silently falling back to a weaker mode.

Profile Resolution

The exact configuration precedence is an application contract. Quecto's flavor resolution is implemented by `resolve` and related functions in `quecto-agent/src/flavor.rs`; read those before relying on a presumed filesystem path or default.

Quecto resolves ordered layers: user base flavor, optional user named flavor, project base flavor, and optional project named flavor. Later layers override keys left unspecified by earlier layers. Do not assume every merge policy is automatically restrictive; trust checks specifically gate project settings that grant privilege.

System invariant: A repository profile *can only restrict*, never *relax*, the user profile. If the user profile says "no sudo" and the repository profile says "allow sudo", the effective result is "no sudo" (the *more restrictive* wins). This is the *default-deny* principle: the user's intent is always honored, even if the repository's intent conflicts.

The Profile File Format

A profile file is a JSON document:

```
{
  "name": "docker-build",
  "allowed_tools": ["ReadFile", "WriteFile", "RunCommand", "Grep"],
  "blocked_commands": ["sudo", "rm -rf /", "git push"],
  "approval_required": [],
  "sub_profiles": {
    "run_docker": {
      "allowed_tools": ["RunCommand"],
      "blocked_commands": ["sudo"],
      "approval_required": ["docker run --privileged"]
    }
  }
}
```

- `name` is the profile's identifier, selected by the application.
- `allowed_tools` is a set (subset of all tools).

- `blocked_commands` is a list of *patterns* (substring match).
- `approval_required` is a list of (tool, path) pairs, or wildcard `*` for *all* paths.
- `sub_profiles` is a map of *named overrides* (resolved dynamically when the agent explicitly requests the sub-profile, via a tool call `use_profile(name)`).

Quecto in production: Quecto flavors are TOML manifests with layered configuration. The `Flavor` type denies unknown fields for its strict sections; `resolve` and related functions combine configuration sources. Project-scoped privilege is subject to trust checks. There is no `quecto profile list/create` subcommand; inspect `quecto-agent --help` and `quecto-agent/src/flavor.rs` for the actual interface and precedence.

Profiles in Production

Quecto uses the `--flavor` option to select named TOML layers. Base files are `~/.config/quecto/flavor.toml` and `<repo>/.quecto/flavor.toml`; optional named files are stored under `flavors/<name>.toml` within those directories. The project layer follows the user layer. Trust-on-first-use protects project flavor settings that grant additional privilege.

Failure mode: If the user override is persisted *globally* (applied to all future tasks), a one-time override becomes permanent. Overrides must *expire* (TTL). The harness should store an override as `{ profile: "full", expires: "2026-10-02T23:59:00Z" }`. After `expires`, the override is *automatically removed*.

The Profile Gate

The policy checks against the active profile:

1. *Is the tool in `allowed_tools`?* (If not, `Deny("not in profile")`).
2. *Does the command match `blocked_commands`?* (If so, `Deny("blocked by profile")`).
3. *Is the (tool, path) pair in `approval_required`?* (If so, `Ask("approval required")`).

If the call passes all three checks, `Allow`. Otherwise, the harness *rejects* it (no execution, no approval prompt — the denial is logged and the tool call is *dropped*).

System invariant: The policy must *never* execute a tool that is denied. The harness must *log* the denial (with the reason) and *skip* the tool call. The agent's next turn will receive a *denial message* (the text from `Deny(String)`), which the model will use to adjust its next call.

Exercise

Write a test that:

- 1. Creates a `docker-build` profile with `allowed_tools: ["RunCommand"]`, `blocked_commands: ["sudo"]`.
- 2. Verifies that `RunCommand("docker build .")` returns `Allow`.
- 3. Verifies that `RunCommand("sudo docker run ...")` returns `Deny("blocked by profile")`.
- 4. Verifies that `WriteFile("main.rs", "...")` returns `Deny("not in profile")`.

This test demonstrates the *profile gate*: a named configuration that *restricts* tools and commands beyond the three presets.

Beyond Rust

CONCERN	PYTHON	TYPESCRIPT
Profile JSON	<code>json.load(f)</code>	<code>JSON.parse(fs.readFileSync(...))</code>
Resolution	<code>resolve_profile(user, repo)</code>	<code>resolveProfile(userProfile, repoProfile)</code>
<code>allowed_tools</code> check	<code>tool_name in allowed</code>	<code>allowedTools.includes(toolName)</code>

Build checkpoint

```
cd books/building-a-coding-agent-harness/examples/reference-harness
cargo test --test safety policy_presets_gate_reads_edits_and_commands
```

This checkpoint exercises the implemented teaching policy. The profile exercise above is a design task, not an existing test.

10. MCP — Model Context Protocol for Tool Expansion

Chapter 10 studies how to expand a harness's tool surface through MCP. The compact reference crate does not implement MCP; production Quecto has a separate `quecto-mcp` crate and an optional adapter in `quecto-agent`.

The Problem

The three built-in tools (ReadFile, WriteFile, ApplyPatch) + RunCommand are *hard-coded* in the harness. If a harness user wants to add a tool (e.g., "run `cargo test` with a custom filter"), they must *modify the harness source* and *recompile*. This defeats the purpose of a *reusable* harness.

MCP (Model Context Protocol) solves this: the harness *discovers* tools at startup (a list of tool schemas), and *executes* tools over a protocol (a request-response pattern). The tool's source lives in a *separate process* (an MCP server). The harness *does not know* the tool's implementation; it only knows the tool's *schema* (name, description, parameters).

System invariant: The harness must *never* load or *execute* a tool without the user's explicit consent (the user must *opt in* to each MCP server). The harness must *not* run MCP servers from untrusted sources (no hash verification of the server's binary). The user *knows* which server is running (from the command line or a trusted path).

The MCP Server

An MCP server is a *process* that:

- Exposes a list of tools (name, description, JSON schema).
- Listens for tool calls (a request with `method`, `params`).
- Returns a result (a response with `result` and `error`).

The server can communicate over:

- **stdio** — the harness starts the process, sends JSON-RPC 2.0 messages over stdin/stdout.
- **HTTP** — the harness POSTs JSON-RPC 2.0 messages to an HTTP endpoint (more complex, but supports remote servers).

The harness *discovers* tools (at startup) by *sending an initialize request* to the server:

```
{
  "jsonrpc": "2.0",
  "method": "initialize",
  "params": { "protocolVersion": "2024-11-05" },
  "id": 1
}
```

The server responds with `tools/list` (an array of tool schemas). The harness *registers* each tool in the `ToolRegistry` (so the model sees them).

Failure mode: If the MCP server *times out* (no initialize response), the harness must *skip* the server (not crash). The harness should log a warning: "MCP server `<name>` did not respond; skipping." The harness *does not block* on a dead MCP server.

The MCP Tool

An MCP tool is a *tool* that the harness executes over the protocol:

- The harness *sends* a request: `{"method": "tools/call", "params": {"name": "read_file", "arguments": {"path": "src/main.rs"}}}`.
- The server *responds* with `{"result": {"content": [...]}}` (or `{"error": {"code": ..., "message": ...}}`).
- The harness *parses* the response (converting it to a `ToolResult`).

The harness *does not care* what the server does. The server might read a file (the built-in `ReadFile`), or run a docker command, or query an API. The harness *only* cares about the schema and the result.

System invariant: The harness must *validate* the MCP response (the response must be a valid JSON-RPC 2.0 object). If the server returns invalid JSON, or an unexpected field, the harness must *not* crash. Instead, it must return a `ToolResult::Failed` with the error message from the server.

The Protocol (JSON-RPC 2.0)

MCP uses JSON-RPC 2.0 over a transport (stdio or HTTP). The protocol defines:

- `initialize` — client sends capabilities, server responds with capabilities + tool list.
- `tools/list` — client requests tool list (called once at startup).
- `tools/call` — client calls a tool (per agent step).
- `notifications/initialized` — client sends an acknowledgment (after initialization).
- `notifications/message` — server sends a log message (for observability).

The harness *must implement* the `initialize` + `tools/list` handshake (at startup) and the `tools/call` execution (per tool call).

Quecto in production: MCP support is optional (`--features mcp`) and provided by the separate `quecto-mcp` crate. The CLI accepts configured STDIO, Streamable HTTP, and legacy SSE server connections. These are distinct transports; do not describe HTTP as an in-process server or assume transport isolation without checking the selected transport and its lifecycle.

The Discovery Protocol

The discovery process is:

1. The harness *starts* the MCP server (as a subprocess, with a known binary path).
2. The harness *sends* the `initialize` request (JSON-RPC 2.0 over stdio).
3. The harness *parses* the response (tool list).
4. The harness *registers* each tool in the `ToolRegistry` (adds the tool name + schema).
5. The harness *sends* `notifications/initialized`.

If the server *fails to respond* (timeout), the harness *logs a warning* and *continues* (without the server's tools). The harness *does not fail* if a server is unreachable.

Failure mode: If the harness *crashes* during initialization (e.g., the server's response is a malformed JSON), the harness *must recover* (log the error, skip the server, continue). The harness *must not* leave the agent in a crashed state.

Exercise

Write a test that:

1. Starts a *mock* MCP server (a test double that returns a fixed tool list).
2. Verifies that the harness registers 2 tools (one named "read_file", one named "write_file").
3. Verifies that the harness *can execute* a tool call (sends a `tools/call`, receives a result).

This test demonstrates the *discovery protocol*: the harness discovers tools (at startup) and *executes* them (over the protocol) without knowing the implementation.

Beyond Rust

CONCERN	PYTHON	TYPESCRIPT
MCP studio	<code>subprocess.Popen</code> + <code>json.loads</code>	<code>child_process.spawn</code> + <code>JSON.parse</code>
Tool list	<code>{"jsonrpc":"2.0","method":"initialize"}</code>	Same
Tool call	<code>{ "method":"tools/call", "params":{"...} }</code>	Same

Build checkpoint

```
cd books/building-a-coding-agent-harness/examples/reference-harness
cd /path/to/quecto
cargo test -p quecto-mcp
```

The MCP client crate's tests exercise its protocol and transports. The mock discovery test described above is a design exercise, not a test in the book's teaching crate.

11. Observability and Telemetry

Chapter 11 studies how to observe an agent's runtime: what it did, when it did it, and where it failed. The JSON Lines API below is a design sketch; this chapter does not add a telemetry module to the runnable teaching crate.

The Problem

A coding agent *succeeds or fails* based on the model's output. But *why* did it fail? The model's reasoning is *internal* (a hidden chain-of-thought, or a unreasoned answer). The harness must *observe* the agent's *visible* behavior: tool calls, file changes, verification results, errors, and the final outcome.

If the harness *does not observe* these, the user (or the harness UI) cannot *diagnose* failures. The user sees: "the agent failed" — but has *no data* about what happened.

A telemetry system is *observability*: it records every visible event (tool calls, results, errors) in a *structured log*. The log is *durable* (written to a file, not in memory). The user (or the harness) can *query* the log (for debugging, for analytics, for auditing).

Design invariant: Decide explicitly whether prompts, tool arguments, results, or model reasoning may be persisted. The JSONL example below excludes prompt and reasoning content and truncates tool data; this is a proposed privacy policy, not a guarantee about Quecto's SQLite transcript storage.

The Telemetry Struct

Implementation boundary: The teaching crate does not contain a telemetry module. This chapter describes instrumentation requirements and compares them with Quecto's optional OpenTelemetry tracing and SQLite-backed run recorder.

In the illustrative JSONL design, a `Telemetry` object could hold:

- `log_path` — the path to the log file (a `.jsonl` file in the session's output directory).
- `session_id` — the current session's ID (used to correlate events).
- `events` — a *buffered* list of events (flushed to disk at session end).

Each event is a JSON object:

```
{
  "timestamp": "2026-10-02T23:59:00Z",
  "session_id": "abc-123",
  "event_type": "tool_call",
  "tool_name": "write_file",
  "arguments": {"path": "src/main.rs", "content": "fn entry() {}"},
  "result": null
}
```

The event types are:

- `tool_call` — a tool was called (recorded before execution).
- `tool_result` — a tool produced a result (recorded after execution).
- `error` — an error occurred (during tool execution, model transport, or verification).
- `session_start` — the session began.
- `session_end` — the session ended (with `outcome`).

Failure mode: If the log file *cannot be written* (permissions, disk full), the telemetry system must *not crash*. It should *log an error* (to stderr) and *continue* (without telemetry). The harness *does not depend* on telemetry for correctness. Telemetry is *observability only*.

The Telemetry API

METHOD	RETURNS
<code>Telemetry::new(log_path)</code>	A telemetry object (writes to the file).
<code>Telemetry::log_event(event)</code>	Writes a JSON object to the file (append mode).
<code>Telemetry::query_since(start_time)</code>	Returns events from <code>start_time</code> (as a list of JSON objects).
<code>Telemetry::flush()</code>	Flushes the buffer to disk (called at session end).

An implementation might call `log_event` at:

- Before every tool execution (a `tool_call` event).
- After every tool execution (a `tool_result` event).
- On every error (an `error` event).
- At session start/end (a `session_start` / `session_end` event).

System invariant: The telemetry must *truncate* arguments and results to 1 KB (to prevent logs from growing to millions of lines). The truncated value is recorded as `"<truncated: 1024 chars>"` (with the full string in a *detailed log file*, if the harness is configured for debug mode).

The Telemetry File Format

The log file is *JSON Lines* (one JSON object per line). Each line is *independent* (the file can be truncated at any line boundary without corruption). The format is:

```
{ "timestamp": "...", "session_id": "...", "event_type": "session_start", "args": {} }
{ "timestamp": "...", "session_id": "...", "event_type": "tool_call", "tool_name": "write_file", .
.. }
{ "timestamp": "...", "session_id": "...", "event_type": "tool_result", "tool_name": "write_file", "result":
{ ... } }
{ "timestamp": "...", "session_id": "...", "event_type": "session_end", "outcome": "Complete", ... }
```

An application using this proposed format could query the file with a JSONL reader:

- `grep -c tool_call .q/.logs/*.jsonl` — total tool calls across all sessions.
- `grep -c error .q/.logs/*.jsonl` — total errors (across all sessions).
- `grep session_start .q/.logs/*.jsonl | wc -l` — number of sessions (concurrent vs sequential).

Quecto in production: Run transcripts and file changes are persisted by `quecto-agent/src/session.rs` and `recorder.rs` in SQLite. Optional OpenTelemetry tracing is initialized from the `otel` feature in `quecto-agent/src/main.rs`; it is exported through OTLP. The CLI has no `quecto logs` command and does not write the JSONL log format sketched above.

The Telemetry as a Debug Tool

Telemetry is *the primary debug tool* for a coding agent. The user (or the harness) can:

1. *Inspect* a session's log (for a specific task).
2. *Compare* two sessions (did the agent fix the same bug twice?).
3. *Analyze* the model's *error rate* (how often does the model call `WriteFile` on a file that already exists? — a redundant call).
4. *Audit* the agent's *approvals* (how often does the agent request approval, and what does the user approve/deny?).

Failure mode: If the telemetry *format* changes (e.g., a new field is added, or a field is removed), the log query (`grep`) must *still work* (backward-compatible queries). The telemetry format *must not* break existing queries. A new field should be *optional* (with a default value).

Exercise

Write a test that:

1. Creates a `Telemetry` object with `log_path = temp_dir() / "test.jsonl"`.
2. Logs 3 events (a `session_start`, a `tool_call` (`write_file`), and a `session_end`).
3. Verifies that the log file contains exactly 3 JSON objects (3 lines).
4. Verifies that `query_since(start_time)` returns 2 events (the `tool_call` and `session_end`).

This is a design exercise for a JSONL telemetry implementation. Quecto uses SQLite for persisted sessions and optional OTLP tracing rather than this proposed file/query API.

Beyond Rust

CONCERN	PYTHON	TYPESCRIPT
JSONL write	<code>json.dump(obj, f)</code>	<code>fs.appendFileSync(file, obj + "\n")</code>
Query	<code>grep -c event file.jsonl</code>	Same (shell)
Session start/end	<code>{event_type: "session_start"}</code>	Same

Build checkpoint

```
cd /path/to/quecto
cargo test -p quecto-agent recorder_appends_messages_and_changes_with_sequence
```

This Quecto test checks that the production recorder persists messages and file changes with sequence numbers. It does not test JSONL output.

12. Evaluation and Benchmarks

Chapter 12 develops an evaluation design for agent harnesses. The simplified `Benchmark` and `Suite` examples are teaching sketches; production Quecto evaluation is a separate crate with its own contract and storage model.

The Problem

The model's quality is *ambiguous* (does it "fix the bug"?). A human says "yes," but the benchmark says "no" (the test output says "3 tests failed"). The harness must *measure* quality *objectively* (verification output, not the human's intuition).

An *evaluation* (or *benchmark*) is a *structured task*:

- A *repository* (the codebase to edit).
- A *prompt* (the user's request: "fix the auth bug").
- A *verification command* (e.g., `cargo test`).
- A *success criterion* (what counts as "pass"?).

A *benchmark suite* is a *collection* of benchmarks (to compare models). The harness *runs* each benchmark (against each model), *collects* the results (pass/fail, steps, time), and *reports* a *score* (percentage passed, average steps).

System invariant: The benchmark *must* be *reproducible*. If the benchmark is *not* reproducible (e.g., the verification depends on a random seed, or the model's output changes per run), the benchmark *is not a benchmark* (it is an *experiment*). The harness *must* record the *seed* (if any) and *pin* the repository state (a commit hash, not a branch name).

The Benchmark Struct

Implementation boundary: Evaluation is discussed as a system design and is not implemented in the compact reference crate. Production evaluation lives in the separate `quecto-eval` crate.

In the simplified sketch, a `Benchmark` could hold:

- `id` — a unique identifier (e.g., "auth-fix-01").
- `repo_url` — the repository URL (or local path).
- `repo_commit` — the repository commit (pinned state).

- `prompt` — the user's prompt (the task description).
- `verification` — the verification commands (a list of strings).
- `pass_threshold` — the pass criteria (e.g., "0 tests failed").
- `expected_steps` — an *expected* step range (for a sanity check: "this benchmark should take 1–5 steps").

This book's simplified benchmark design uses a manifest to pin task inputs. Quecto's production evaluator uses its own manifest and contract types; do not assume the illustrative YAML/JSON shape is accepted by `quecto-eval`.

Failure mode: If the repository *clone* fails (network down, repository deleted), the benchmark *must* record a `Skipped` result (not a crash). The harness *does not fail* the entire suite because one benchmark is skipped.

The Benchmark Runner

An isolated benchmark runner should run each task against a fixed repository state:

1. *Clone* the repository at `repo_commit`.
2. *Run* the agent (with the prompt, the verification commands).
3. *Collect* the result (pass/fail, steps, time).
4. *Record* the result (to a JSON file).

The harness *does not modify* the original repository (it works in a *temporary clone*). After the benchmark runs, the harness *deletes* the clone (to avoid disk space leaks).

System invariant: The benchmark runner *must isolate* each benchmark (a benchmark's file changes *must not* affect the next benchmark). The harness *must delete* the cloned repository after each benchmark.

The Benchmark Suite

A *suite* is a collection of benchmarks (with metadata). The suite file is:

```
name: "coding-agent-harness-benchmarks"
version: "1.0"
benchmarks:
  - "auth-fix-01"
  - "refactor-api-02"
  - "test-cleanup-03"
  - "feature-add-04"
```

The harness *loads* the suite (reads the file), *resolves* each benchmark (loads the JSON file), and *runs* them (sequentially, or in parallel — with a configurable concurrency limit).

The harness *reports* a suite score:

- `total` — number of benchmarks.
- `passed` — number of benchmarks that passed.
- `failed` — number of benchmarks that failed.
- `skipped` — number of benchmarks that were skipped (failed to clone, timed out).
- `avg_steps` — average step count across *passed* benchmarks.

Quecto in production: Evaluation is a separate workspace member with `eval` and `compat` CLI subcommands (`cargo run -p quecto-eval -- --help`). The compatibility runner accepts a manifest, tasks directory, database path, and agent binary. It is not a `quecto benchmark` subcommand and does not emit the invented score schema shown in early drafts. Read `quecto-eval/src/cli.rs`, `manifest.rs`, and `runner.rs` for the current contract.

The Score Format

The score file is a JSON object:

```
{
  "suite_name": "coding-agent-harness-benchmarks",
  "runs_at": "2026-10-02T23:59:00Z",
  "results": {
    "auth-fix-01": {
      "outcome": "Complete",
      "steps": 3,
      "time_ms": 45000,
      "verification_output": "2/2 tests passed"
    }
  },
  "score": {
    "passed": 4,
    "failed": 0,
    "skipped": 0,
    "total": 4,
    "avg_steps": 3.0
  }
}
```

The harness *reports* the score (to the user or the CLI output). The user *can compare* two score files (e.g., "Model A: avg 3.0 steps, 4/4 passed; Model B: avg 5.2 steps, 3/4 passed").

Failure mode: If the score file *is corrupted* (an incomplete JSON), the harness *must not* crash. It should *log a warning* (and skip that score file). The harness *does not* use a corrupted score for comparison.

Exercise

Write a test that:

- 1. Creates a benchmark suite with 2 benchmarks (a "pass" and a "fail" benchmark).
- 2. Runs the suite (with a *mocked* agent that returns "Complete" for the pass benchmark and "VerificationFailed" for the fail benchmark).
- 3. Verifies that the score file has `passed: 1, failed: 1, total: 2`.
- 4. Verifies that `avg_steps` is a valid number (not null).

This is a design exercise for the simplified suite API described here; the production evaluator uses its own contracts and result storage.

Beyond Rust

CONCERN	PYTHON	TYPESCRIPT
Benchmark file	<code>yaml.safe_load(f)</code>	<code>JSON.parse(fs.readFileSync(...))</code>
Suite score	<code>{passed: 4, total: 4}</code>	Same
Session clone	<code>git clone <url> --depth 1</code>	Same (shell)

Build checkpoint

```
cd /path/to/quecto
cargo test -p quecto-eval
```

This runs the production evaluator's unit and integration tests. The compact book crate has no `evaluate` test target.

13. The Complete Reference Harness

This chapter assembles the runnable teaching crate from Chapters 1–5 and connects it to the runtime concerns studied in Chapters 6–12. The book's crate is intentionally compact; it is not a duplicate of production Quecto.

What You Have Built

At this point, the reference crate has a real model boundary, a bounded loop, typed tools, repository path checks, a command boundary, and a policy gate. Its structure is deliberately small:

```
reference-harness/
├── Cargo.toml
├── src/
│   ├── lib.rs          # module exports
│   ├── model.rs        # messages, Model, OpenAI-compatible HTTP
│   ├── agent.rs        # loop, outcomes, repeat/denial limits
│   ├── tools.rs        # typed tools and deterministic registry
│   ├── context.rs      # repository paths and bounded command runner
│   └── policy.rs       # allow / ask / deny decisions
└── tests/
    ├── agent_loop.rs
    ├── model_http.rs
    └── safety.rs
```

The implementation does **not** include production sessions, profiles, verification retries, MCP, telemetry, evaluation, or a CLI. Those concerns are explored as design extensions and mapped to production source in Appendix A. Keeping that distinction visible matters: a reader should be able to run every claim about the teaching crate without discovering that the corresponding file was never written.

Exercise the Integrated Core

Run the full reference suite:

```
cd books/building-a-coding-agent-harness/examples/reference-harness
cargo test
```

The integration tests use a scripted model and local mock HTTP server, so they do not need model credentials. They exercise normal completion, tool dispatch, bounded stopping, repeat detection, policy decisions, repository path containment, and HTTP error handling. Tests that spawn shell processes are marked ignored in restricted environments; run them only where subprocess creation is permitted.

The crate is a library component, not a ready-to-use coding-agent executable. To turn it into one, add a small CLI that parses a repository path and task, constructs `Context`, registers the built-in tools, chooses a policy preset, builds `HttpModel` from environment configuration, and invokes `Agent::run`. Keep the CLI as an adapter: transport, policy, and tool behavior should remain testable without terminal input.

Production Quecto Is the Larger Reference

The production system is a Cargo workspace rather than this one-crate teaching example:

```

quecto/           # synchronous OpenAI-compatible core
quecto-agent/     # coding-agent runtime and CLI
quecto-mcp/       # optional MCP client transports
quecto-eval/      # evaluation and compatibility experiments

```

The `quecto-agent` library exports its public agent, model, policy, tool, sandbox, session, verification, flavor, and trust APIs. Its binary is named `quecto-agent`; it supports direct prompts and the `chat`, `resume`, `undo`, `diff`, and `new` subcommands. MCP and OpenTelemetry are opt-in Cargo features. Consult `quecto-agent --help` and Appendix A instead of relying on commands from an earlier draft of this book.

From a Quecto checkout, the general verification entry point is:

```
cargo test --workspace
```

The teaching crate and production workspace intentionally have different public types. Port a concept, not an assumed API: compare the invariant first, then identify how the production system enforces it.

Design Review Checklist

Before adapting the harness to a real repository, answer these questions:

1. Is every model response represented as either ordinary text or a typed tool call?
2. Are tool names, arguments, paths, and command permissions checked before side effects?
3. Are loop limits and error outcomes explicit and testable?
4. Can a user distinguish a model claim from a verified result?
5. Are secrets excluded from logs and diagnostics by construction?
6. Can an interrupted task be recovered without replaying unsafe operations?
7. Can an evaluation reproduce the agent, tool set, repository revision, and verification commands used?

These are not all implemented in the book's compact crate. Treat unanswered items as deliberate follow-up engineering, not as implied features.

End of Chapter 13.

Appendix A. The Quecto Reference

Quecto is the production system behind this book's examples. The repository is at <https://github.com/adityak74/quecto>. The chapter mapping below uses the local reference checkout at revision `97158860a490790edeb58f9727b684423f04cbe3` (2026-10-03). Paths are relative to that repository root. Check the remote `main` branch for newer source before treating a path as current.

The Runnable Book Harness

The compact, self-contained teaching crate is `books/building-a-coding-agent-harness/examples/reference-harness`. It has five modules: `model`, `agent`, `tools`, `context`, and `policy`. This crate is deliberately smaller than Quecto. The chapters about persistence, profile composition, MCP, telemetry, and evaluation explain production concerns and map them to Quecto; they do not claim those systems are implemented in this teaching crate.

BOOK TOPIC	TEACHING IMPLEMENTATION	PRODUCTION QUECTO SOURCE
Model messages and HTTP	<code>examples/reference-harness/src/model.rs</code>	<code>quecto-agent/src/model.rs</code> , plus root <code>src/lib.rs</code> for the tiny core
Bounded loop and outcomes	<code>examples/reference-harness/src/agent.rs</code>	<code>quecto-agent/src/agent.rs</code>
Tools and schemas	<code>examples/reference-harness/src/tools.rs</code>	<code>quecto-agent/src/tools/</code> and <code>quecto-agent/src/lib.rs</code>
Repository context and command sandbox	<code>examples/reference-harness/src/context.rs</code>	<code>quecto-agent/src/context.rs</code> , <code>quecto-agent/src/sandbox.rs</code>
Execution policy	<code>examples/reference-harness/src/policy.rs</code>	<code>quecto-agent/src/policy.rs</code> , <code>quecto-agent/src/approval.rs</code>
Verification	Design chapter; not in the teaching crate	<code>quecto-agent/src/verify.rs</code>
Instructions	Design chapter; not in the teaching crate	<code>quecto-agent/src/instructions.rs</code>
Sessions and persistence	Design chapter; not in the teaching crate	<code>quecto-agent/src/session.rs</code> , <code>quecto-agent/src/recorder.rs</code>
Profiles / layered config	Design chapter; not in the teaching crate	<code>quecto-agent/src/flavor.rs</code>
MCP	Design chapter; not in the teaching crate	<code>quecto-mcp/</code> and feature-gated <code>quecto-agent/src/mcp_adapter.rs</code>
OpenTelemetry	Design chapter; not in the teaching crate	<code>quecto-agent/src/main.rs</code> (<code>otel</code> feature), <code>quecto-agent/Cargo.toml</code>

BOOK TOPIC	TEACHING IMPLEMENTATION	PRODUCTION QUECTO SOURCE
Evaluation	Design chapter; not in the teaching crate	<code>quecto-eval/</code>

How to Explore the Production Repository

Quecto is a Cargo workspace. Its root `quecto` crate is the small synchronous OpenAI-compatible model transport. `quecto-agent` builds the interactive and one-shot coding agent on top of it; `quecto-mcp` contains MCP transport support; `quecto-eval` contains evaluation contracts, manifests, and runners. Begin with each crate's `src/lib.rs` and `Cargo.toml`, then follow the named module paths in the table.

Useful commands from a Quecto checkout:

```
cargo test --workspace
cargo run -p quecto-agent -- --help
cargo run -p quecto-agent -- "Summarize this repository"
cargo run -p quecto-eval -- --help
```

Optional features alter the build: MCP is enabled with `--features mcp`; OpenTelemetry with `--features otel`. The default agent build does not enable these optional features.

What the Two Implementations Share

CONCERN	TEACHING HARNESS	PRODUCTION QUECTO
Model boundary	Small <code>Model::complete</code> trait and OpenAI-compatible HTTP implementation	Core model transport plus the agent crate's own model abstraction and provider wire-format handling
Agent	Bounded synchronous loop with step, repeat, and policy-denial limits	Coding-agent run loop with model completion options, tools, session data, and recorder hooks
Tools	Read, write, patch, and command examples	File, search, git, shell, and optional MCP-backed tools
Policy	Three educational presets and hard-deney examples	Approval modes, flavors, tool filtering, and trust configuration
Verification	Architectural treatment only	Configured completion gate implemented by <code>quecto-agent/src/verify.rs</code>
State	In-memory teaching objects	SQLite-backed session storage and change summaries
Telemetry	Not implemented in the teaching crate	Optional OpenTelemetry tracing; do not confuse it with a JSONL session log

CONCERN	TEACHING HARNESS	PRODUCTION QUECTO
Evaluation	Not implemented in the teaching crate	Separate <code>quecto-eval</code> crate and evaluation data

The teaching code prioritizes readable boundaries over feature parity. It is not a drop-in library for production use. Production Quecto has its own types, error paths, config system, and platform constraints; compare behaviors rather than assuming identical APIs.

CLI Orientation

The production binary is `quecto-agent`, not a `quecto run` subcommand. It accepts a task prompt directly, supports `chat`, `resume`, `undo`, `diff`, and `new` subcommands, and provides flags such as `--yes`, `--no-verify`, `--flavor`, `--model`, `--base-url`, and `--max-steps`. Run `cargo run -p quecto-agent -- --help` for the current interface. Evaluation is a separate binary in `quecto-eval`; it is not a `quecto benchmark` subcommand.

Further Reading

- 1. [Quecto repository](#) — source, README, and release information.
- 2. [Anthropic Messages API](#) — one provider's message interface.
- 3. [OpenAI function calling](#) — tool-call schemas and request flow.
- 4. [JSON-RPC 2.0 specification](#) — protocol foundations relevant to MCP.
- 5. [Cargo workspaces](#) — how the production crates fit together.

Glossary

TERM	DEFINITION
Harness	The runtime around a model that controls tools, state, policy, and completion.
Policy	A gate that allows, asks about, or denies an action before execution.
Session	The persisted state for an agent run, including messages and recoverable changes.
Flavor	Quecto's layered configuration profile for model, tools, approvals, and verification.
MCP	Model Context Protocol, used to discover and invoke external tools through servers.
Telemetry	Instrumentation emitted for operational observation; in Quecto this means optional OpenTelemetry, not a local JSONL CLI log.
Evaluation	Repeatable task execution with defined inputs and measurable outcomes.

Rebuild This Book

The source manuscript, build script, stylesheet, and reference crate live together in the book directory. From `books/building-a-coding-agent-harness/examples`, run:

```
python3 build_book.py
cargo test --manifest-path reference-harness/Cargo.toml
```

The builder writes the PDF and HTML to `static/books/`. It resolves source-code includes from the reference crate so the short listings stay aligned with executable code.

End of Appendix A.